$$\begin{array}{llllll} 1 & a_2 & a_4 & a_6 & \dots & a_n \\ a_1 & a_3 & a_5 & \dots & & \\ b_1 & b_2 & b_3 & \dots & & \\ c_1 & c_2 & \cdot & & & \\ \cdot & \cdot & \cdot & & & \\ \cdot & \cdot & \cdot & & & \\ \cdot & \cdot & \cdot & & & \end{array}$$

The first term in the first column of the Routh array is 1. The next term in the first column is a_1, which is equal to Δ_1. The next term is b_1, which is equal to

$$\frac{a_1 a_2 - a_3}{a_1} = \frac{\Delta_2}{\Delta_1}$$

The next term in the first column is c_1, which is equal to

$$\begin{aligned} \frac{b_1 a_3 - a_1 b_2}{b_1} &= \frac{\left[\dfrac{a_1 a_2 - a_3}{a_1}\right] a_3 - a_1\left[\dfrac{a_1 a_4 - a_5}{a_1}\right]}{\left[\dfrac{a_1 a_2 - a_3}{a_1}\right]} \\ &= \frac{a_1 a_2 a_3 - a_3^2 - a_1^2 a_4 + a_1 a_5}{a_1 a_2 - a_3} \\ &= \frac{\Delta_3}{\Delta_2} \end{aligned}$$

In a similar manner the remaining terms in the first column of the Routh array can be found.

The Routh array has the property that the last nonzero terms of any columns are the same; that is, if the array is given by

$$\begin{array}{llll} a_0 & a_2 & a_4 & a_6 \\ a_1 & a_3 & a_5 & a_7 \\ b_1 & b_2 & b_3 & \\ c_1 & c_2 & c_3 & \\ d_1 & d_2 & & \\ e_1 & e_2 & & \\ f_1 & & & \\ g_1 & & & \end{array}$$

then

$$a_7 = c_3 = e_2 = g_1$$

and if the array is given by

$$\begin{array}{llll} a_0 & a_2 & a_4 & a_6 \\ a_1 & a_3 & a_5 & 0 \\ b_1 & b_2 & b_3 & \\ c_1 & c_2 & 0 & \\ d_1 & d_2 & & \\ e_1 & 0 & & \\ f_1 & & & \end{array}$$

then

$$a_6 = b_3 = d_2 = f_1$$

In any case, the last term of the first column is equal to a_n, or

$$a_n = \frac{\Delta_{n-1}a_n}{\Delta_{n-1}} = \frac{\Delta_n}{\Delta_{n-1}}$$

For example, if $n = 4$, then

$$\Delta_4 = \begin{vmatrix} a_1 & 1 & 0 & 0 \\ a_3 & a_2 & a_1 & 1 \\ a_5 & a_4 & a_3 & a_2 \\ a_7 & a_6 & a_5 & a_4 \end{vmatrix} = \begin{vmatrix} a_1 & 1 & 0 & 0 \\ a_3 & a_2 & a_1 & 1 \\ 0 & a_4 & a_3 & a_2 \\ 0 & 0 & 0 & a_4 \end{vmatrix} = \Delta_3 a_4$$

Thus it has been shown that the first column of the Routh array is given by

$$1, \quad \Delta_1, \quad \frac{\Delta_2}{\Delta_1}, \quad \frac{\Delta_3}{\Delta_2}, \quad \ldots, \quad \frac{\Delta_n}{\Delta_{n-1}}$$

A-5-12. The value of the gas constant for any gas may be determined from accurate experimental observations of simultaneous values of p, v, and T.

Obtain the gas constant R_{air} for air. Note that at 32°F and 14.7 psia the specific volume of air is 12.39 ft³/lb. Then obtain the capacitance of a 20-ft³ pressure vessel that contains air at 160°F. Assume that the expansion process is isothermal.

Solntion.

$$R_{air} = \frac{pv}{T} = \frac{14.7 \times 144 \times 12.39}{460 + 32} = 53.3 \text{ ft-lb}_f/\text{lb °R}$$

Referring to Equation (5–12), the capacitance of a 20-ft³ pressure vessel is

$$C = \frac{V}{nR_{air}T} = \frac{20}{1 \times 53.3 \times 620} = 6.05 \times 10^{-4} \frac{\text{lb}}{\text{lb}_f/\text{ft}^2}$$

Note that in terms of SI units, R_{air} is given by

$$R_{air} = 287 \text{ N-m/kg K}$$

A-5-13. Figure 5–67 is a schematic diagram of a pneumatic diaphragm valve. At steady state the control pressure from a controller is $\bar{P}_c$, the pressure in the valve is also $\bar{P}_c$, and the valve-stem displacement is $\bar{X}$. Assume that at $t = 0$ the control pressure is changed from $\bar{P}_c$ to $\bar{P}_c + p_c$. Then the valve pressure will be changed from $\bar{P}_c$ to $\bar{P}_c + p_v$. The change in valve pressure p_v will cause the valve-stem displacement to change from $\bar{X}$ to $\bar{X} + x$. Find the transfer function between the change in valve-stem displacement x and the change in control pressure p_c.

Solntion. Let us define the airflow rate to the diaphragm valve through resistance R as q. Then

$$q = \frac{p_c - p_v}{R}$$

For the air chamber in the diaphragm valve, we have

$$C\,dp_v = q\,dt$$

Consequently,

$$C\frac{dp_v}{dt} = q = \frac{p_c - p_v}{R}$$

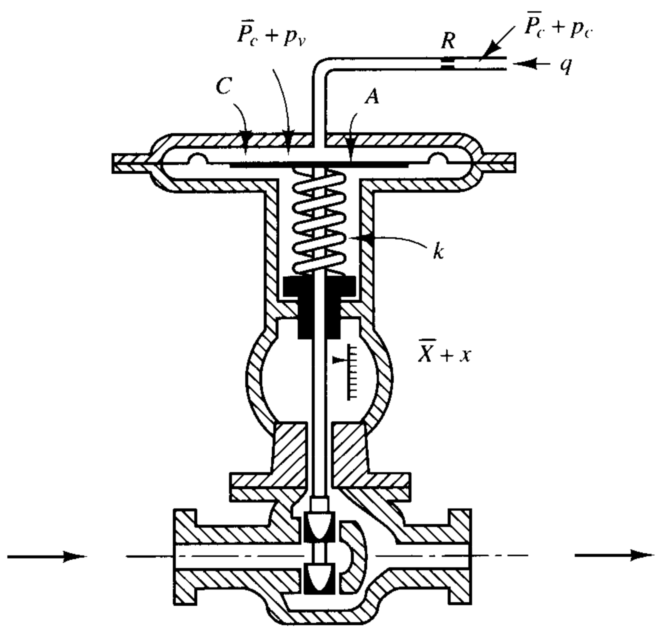


Figure 5–67
Pneumatic diaphragm valve.

from which

$$RC\frac{dp_v}{dt} + p_v = p_c$$

Noting that

$$Ap_v = kx$$

we have

$$\frac{k}{A}\left(RC\frac{dx}{dt} + x\right) = p_c$$

The transfer function between x and p_c is

$$\frac{X(s)}{P_c(s)} = \frac{A/k}{RCs + 1}$$

A-5-14. In the pneumatic pressure system of Figure 5–68(a), assume that, for $t < 0$, the system is at steady state and that the pressure of the entire system is $\bar{P}$. Also, assume that the two bellows are identical. At $t = 0$, the input pressure is changed from $\bar{P}$ to $\bar{P} + p_i$. Then the pressures in bellows 1 and 2 will change from $\bar{P}$ to $\bar{P} + p_1$ and from $\bar{P}$ to $\bar{P} + p_2$, respectively. The capacity (volume) of each bellows is 5×10^{-4} m^3, and the operating pressure difference Δp (difference between p_i and p_1 or difference between p_i and p_2) is between -0.5×10^5 N/m^2 and 0.5×10^5 N/m^2. The corresponding mass flow rates (kg/sec) through the valves are shown in Figure 5–68(b). Assume that the bellows expand or contract linearly with the air pressures applied to them, that the equivalent spring constant of the bellows system is $k = 1 \times 10^5$ N/m, and that each bellows has area $A = 15 \times 10^{-4}$ m^2.

Defining the displacement of the midpoint of the rod that connects two bellows as x, find the transfer function $X(s)/P_i(s)$. Assume that the expansion process is isothermal and that the temperature of the entire system stays at 30°C.

Solution. Referring to Section 5–6, transfer function $P_1(s)/P_i(s)$ can be obtained as

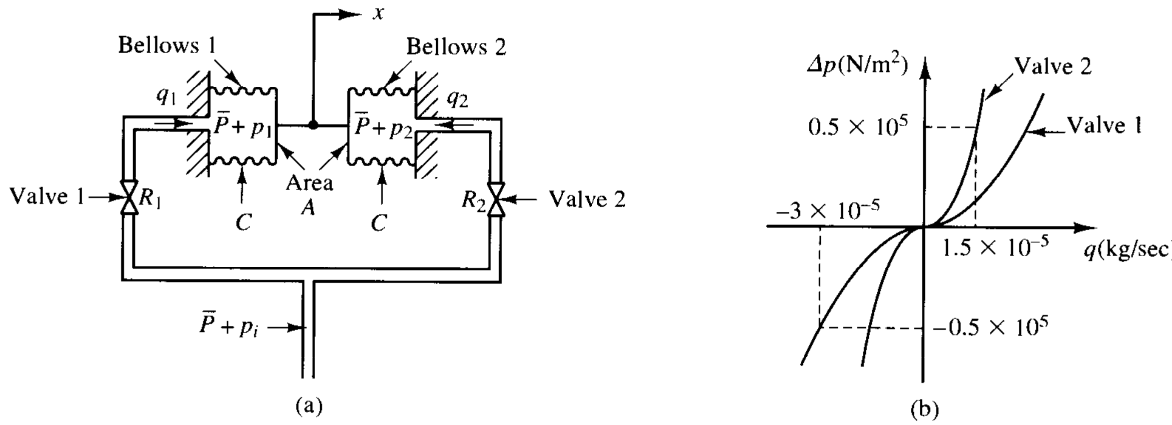


Figure 5–68
(a) Pneumatic pressure system; (b) pressure difference versus mass flow rate curve.

$$\frac{P_1(s)}{P_i(s)} = \frac{1}{R_1Cs + 1} \tag{5–38}$$

Similarly, transfer function $P_2(s)/P_i(s)$ is

$$\frac{P_2(s)}{P_i(s)} = \frac{1}{R_2Cs + 1} \tag{5–39}$$

The force acting on bellows 1 in the x direction is $A(\bar{P} + p_1)$, and the force acting on bellows 2 in the negative x direction is $A(\bar{P} + p_2)$. The resultant force balances with kx, the equivalent spring force of the corrugated side of the bellows.

$$A(p_1 - p_2) = kx$$

or

$$A[P_1(s) - P_2(s)] = kX(s) \tag{5–40}$$

Referring to Equations (5–38) and (5–39), we see that

$$\begin{aligned} P_1(s) - P_2(s) &= \left(\frac{1}{R_1Cs + 1} - \frac{1}{R_2Cs + 1}\right)P_i(s) \\ &= \frac{R_2Cs - R_1Cs}{(R_1Cs + 1)(R_2Cs + 1)}P_i(s) \end{aligned}$$

By substituting this last equation into Equation (5–40) and rewriting, the transfer function $X(s)/P_i(s)$ is obtained as

$$\frac{X(s)}{P_i(s)} = \frac{A}{k}\frac{(R_2C - R_1C)s}{(R_1Cs + 1)(R_2Cs + 1)} \tag{5–41}$$

The numerical values of average resistances R_1 and R_2 are

$$R_1 = \frac{d\,\Delta p}{dq_1} = \frac{0.5 \times 10^5}{3 \times 10^{-5}} = 0.167 \times 10^{10}\,\frac{\text{N/m}^2}{\text{kg/sec}}$$

$$R_2 = \frac{d\,\Delta p}{dq_2} = \frac{0.5 \times 10^5}{1.5 \times 10^{-5}} = 0.333 \times 10^{10}\,\frac{\text{N/m}^2}{\text{kg/sec}}$$

The numerical value of capacitance C of each bellows is

$$C = \frac{V}{nR_{\text{air}}T} = \frac{5 \times 10^{-4}}{1 \times 287 \times (273 + 30)} = 5.75 \times 10^{-9}\,\frac{\text{kg}}{\text{N/m}^2}$$

where $R_{air} = 287$ N-m/kg K. (See Problem A-5-12.) Consequently,

$$R_1C = 0.167 \times 10^{10} \times 5.75 \times 10^{-9} = 9.60 \text{ sec}$$

$$R_2C = 0.333 \times 10^{10} \times 5.75 \times 10^{-9} = 19.2 \text{ sec}$$

By substituting the numerical values for A, k, R_1C, and R_2C into Equation (5–41), we obtain

$$\frac{X(s)}{P_i(s)} = \frac{1.44 \times 10^{-7}s}{(9.6s + 1)(19.2s + 1)}$$

A-5-15. Draw a block diagram of the pneumatic controller shown in Figure 5–69. Then derive the transfer function of this controller.

If the resistance R_d is removed (replaced by the line-sized tubing), what control action do we get? If the resistance R_i is removed (replaced by the line-sized tubing), what control action do we get?

Solution. Let us assume that when $e = 0$ the nozzle–flapper distance is equal to $\bar{X}$ and the control pressure is equal to $\bar{P}_c$. In the present analysis, we shall assume small deviations from the respective reference values as follows:

e = small error signal

x = small change in the nozzle-flapper distance

p_c = small change in the control pressure

p_{I} = small pressure change in bellows I due to small change in the control pressure

p_{II} = small pressure change in bellows II due to small change in the control pressure

y = small displacement at the lower end of the flapper

In this controller, p_c is transmitted to bellows I through the resistance R_d. Similarly, p_c is transmitted to bellows II through the series of resistances R_d and R_i. An approximate relationship between p_{I} and p_c is

$$\frac{P_{\text{I}}(s)}{P_c(s)} = \frac{1}{R_dCs + 1} = \frac{1}{T_ds + 1}$$

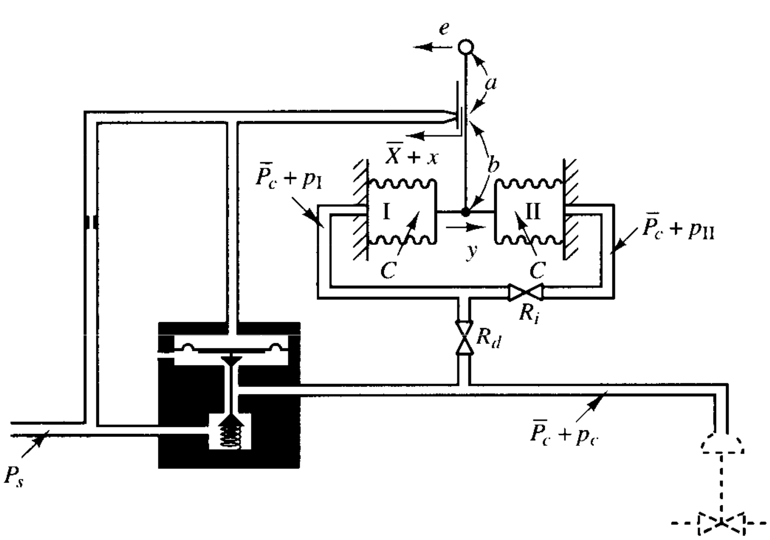


Figure 5–69
Schematic diagram of a pneumatic controller.

where $T_d = R_dC$ = derivative time. Similarly, p_{II} and p_I are related by the transfer function

$$\frac{P_{II}(s)}{P_I(s)} = \frac{1}{R_iCs + 1} = \frac{1}{T_is + 1}$$

where $T_i = R_iC$ = integral time. The force–balance equation for the two bellows is

$$(p_I - p_{II})A = k_s y$$

where k_s is the stiffness of the two connected bellows and A is the cross-sectional area of the bellows. The relationship among the variables e, x, and y is

$$x = \frac{b}{a + b}e - \frac{a}{a + b}y$$

The relationship between p_c and x is

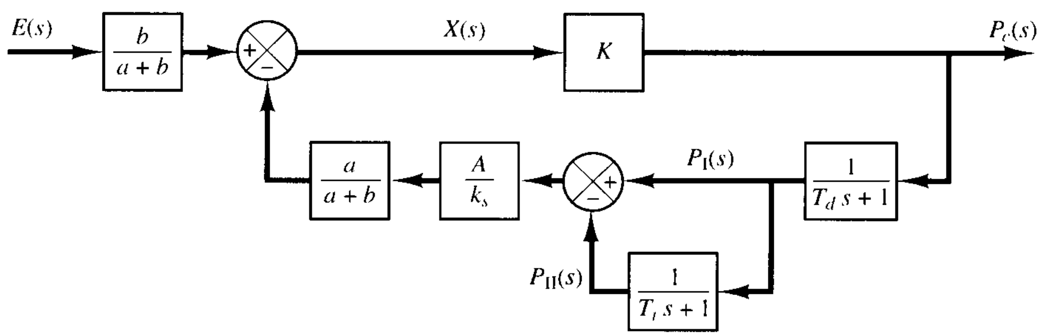
$$p_c = Kx \qquad (K > 0)$$

From the equations just derived, a block diagram of the controller can be drawn, as shown in Figure 5–70(a). Simplification of this block diagram results in Figure 5–70(b).

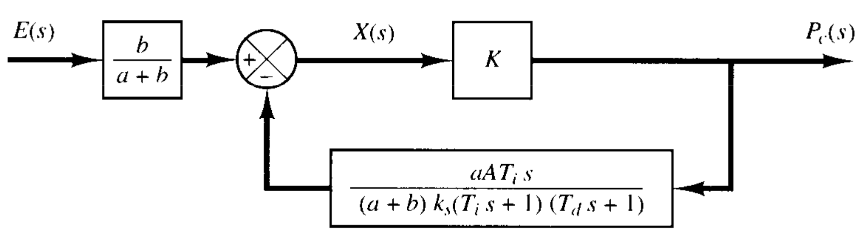
The transfer function between $P_c(s)$ and $E(s)$ is

$$\frac{P_c(s)}{E(s)} = \frac{\frac{b}{a + b}K}{1 + K\frac{a}{a + b}\frac{A}{k_s}\left(\frac{T_is}{T_is + 1}\right)\left(\frac{1}{T_ds + 1}\right)}$$

For a practical controller, under normal operation $KaAT_is/[(a + b)k_s(T_is + 1)(T_ds + 1)]$ is very much greater than unity and $T_i \gg T_d$. Therefore, the transfer function can be simplified as follows:



(a)



(b)

Figure 5–70
(a) Block diagram of the pneumatic controller shown in Figure 5–69; (b) simplified block diagram.

$$\frac{P_c(s)}{E(s)} \doteqdot \frac{bk_s(T_i s + 1)(T_d s + 1)}{aAT_i s}$$
$$= \frac{bk_s}{aA}\left(\frac{T_i + T_d}{T_i} + \frac{1}{T_i s} + T_d s\right)$$
$$\doteqdot K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

where

$$K_p = \frac{bk_s}{aA}$$

Thus the controller shown in Figure 5–69 is a proportional-plus-integral-plus-derivative one.

If the resistance R_d is removed, or $R_d = 0$, the action becomes that of a proportional-plus-integral controller.

If the resistance R_i is removed, or $R_i = 0$, the action becomes that of a narrow-band proportional, or two-position, controller. (Note that the actions of two feedback bellows cancel each other, and there is no feedback.)

A-5-16. Actual spool valves are either overlapped or underlapped because of manufacturing tolerances. Consider the overlapped and underlapped spool valves shown in Figure 5–71(a) and (b). Sketch curves relating the uncovered port area A versus displacement x.

Solution. For the overlapped valve, a dead zone exists between $-\frac{1}{2}x_0$ and $\frac{1}{2}x_0$, or $-\frac{1}{2}x_0 < x < \frac{1}{2}x_0$. The uncovered port area A versus displacement x curve is shown in Figure 5–72(a). Such an overlapped valve is unfit as a control valve.

For the underlapped valve, the port area A versus displacement x curve is shown in Figure 5–72(b). The effective curve for the underlapped region has a higher slope, meaning a higher sensitivity. Valves used for controls are usually underlapped.

A-5-17. Figure 5–73 shows a hydraulic jet-pipe controller. Hydraulic fluid is ejected from the jet pipe. If the jet pipe is shifted to the right from the neutral position, the power piston moves to the left, and vice versa. The jet pipe valve is not used as much as the flapper valve because of large null flow, slower response, and rather unpredictable characteristics. Its main advantage lies in its insensitivity to dirty fluids.

Suppose that the power piston is connected to a light load so that the inertia force of the load element is negligible compared to the hydraulic force developed by the power piston. What type of control action does this controller produce?

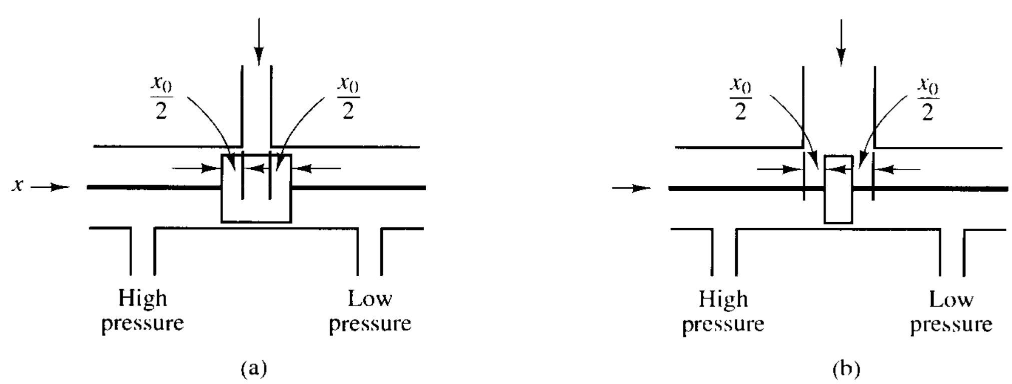


Figure 5–71
(a) Overlapped spool valve; (b) underlapped spool valve.

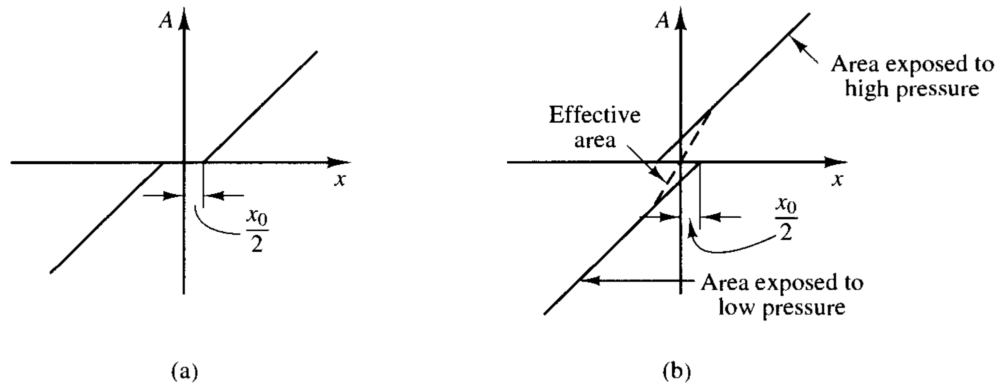


Figure 5–72
(a) Uncovered port area A versus displacement x curve for the overlapped valve; (b) uncovered port area A versus displacement x curve for the underlapped valve.

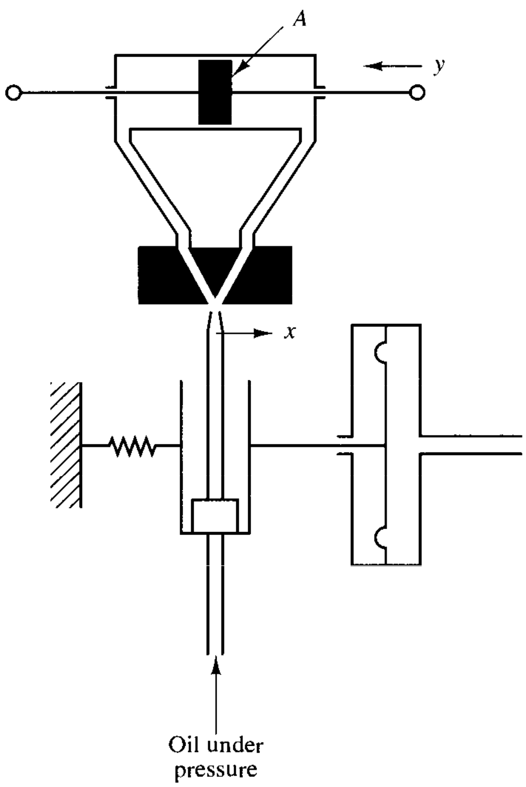


Figure 5–73
Hydraulic jet-pipe controller.

Solution. Define the displacement of the jet nozzle from the neutral position as x and the displacement of the power piston as y. If the jet nozzle is moved to the right by a small displacement x, the oil flows to the right side of the power piston, and the oil in the left side of the power piston is returned to the drain. The oil flowing into the power cylinder is at high pressure; the oil flowing out from the power cylinder into the drain is at low pressure. The resulting pressure difference causes the power piston to move to the left.

For a small jet nozzle displacement x, the flow rate q to the power cylinder is proportional to x; that is,

$$q = K_1 x$$

For the power cylinder,

$$A\rho \, dy = q \, dt$$

where A is the power piston area and ρ is the density of oil. Hence

$$\frac{dy}{dt} = \frac{q}{A\rho} = \frac{K_1}{A\rho}x = Kx$$

where $K = K_1/(A\rho)$ = constant. The transfer function $Y(s)/X(s)$ is thus

$$\frac{Y(s)}{X(s)} = \frac{K}{s}$$

The controller produces the integral control action.

A-5-18. Figure 5–74 shows a hydraulic jet-pipe applied to a flow control system. The jet-pipe controller governs the position of the butterfly valve. Discuss the operation of this system. Plot a possible curve relating the displacement x of the nozzle to the total force F acting on the power piston.

Solution. The operation of this system is as follows: The flow rate is measured by the orifice, and the pressure difference produced by this orifice is transmitted to the diaphragm of the pressure-measuring device. The diaphragm is connected to the free swinging nozzle, or jet pipe, through a linkage. High-pressure oil ejects from the nozzle all the time. When the nozzle is at a neutral position, no oil flows through either of the pipes to move the power piston. If the nozzle is displaced by the motion of the balance arm to one side, the high-pressure oil flows through the corresponding pipe, and the oil in the power cylinder flows back to the sump through the other pipe.

Assume that the system is initially at rest. If the reference input is changed suddenly to a higher flow rate, then the nozzle is moved in such a direction as to move the power piston and open the butterfly valve. Then the flow rate will increase, the pressure difference across the orifice becomes larger, and the nozzle will move back to the neutral position. The movement of the power piston stops when x, the displacement of the nozzle, comes back to and stays at the neutral position. (The jet pipe controller thus possesses an integrating property.)

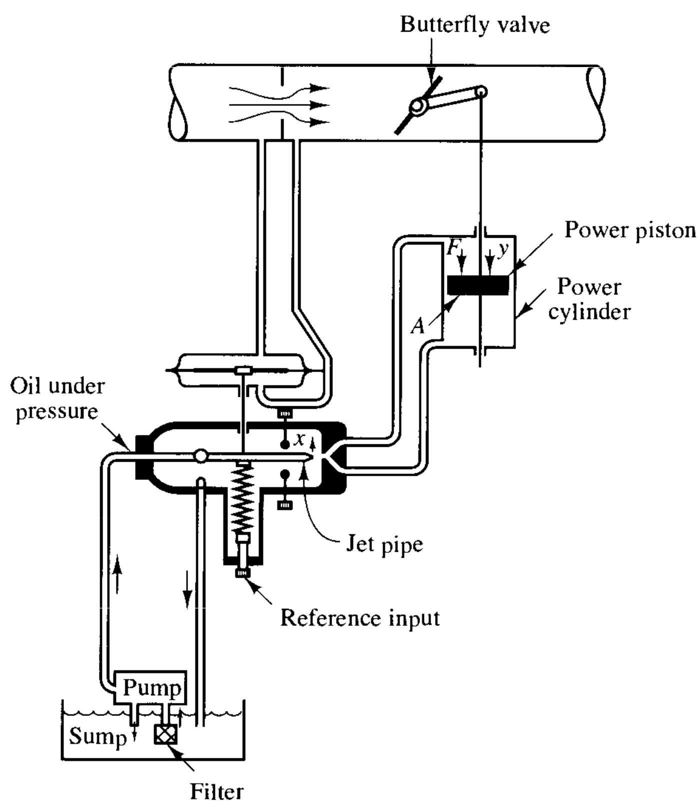


Figure 5–74 Schematic diagram of a flow control system using a hydraulic jet-pipe controller.

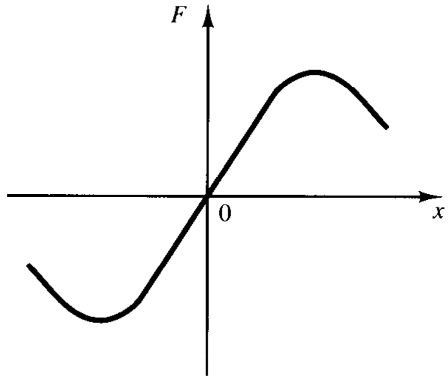


Figure 5–75
Force versus diplacement curve.

The relationship between the total force F acting on the power piston and the displacement x of the nozzle is shown in Figure 5–75. The total force is equal to the pressure difference ΔP across the piston times the area A of the power piston. For a small displacement x of the nozzle, the total force F and displacement x may be considered proportional.

A-5-19. Explain the operation of the speed control system shown in Figure 5–76.

Solution. If the engine speed increases, the sleeve of the fly-ball governor moves upward. This movement acts as the input to the hydraulic controller. A positive error signal (upward motion of the sleeve) causes the power piston to move downward, reduces the fuel-valve opening, and decreases the engine speed. A block diagram for the system is shown in Figure 5–77.

From the block diagram the transfer function $Y(s)/E(s)$ can be obtained as

$$\frac{Y(s)}{E(s)} = \frac{a_2}{a_1 + a_2} \frac{\dfrac{K}{s}}{1 + \dfrac{a_1}{a_1 + a_2} \dfrac{bs}{bs + k} \dfrac{K}{s}}$$

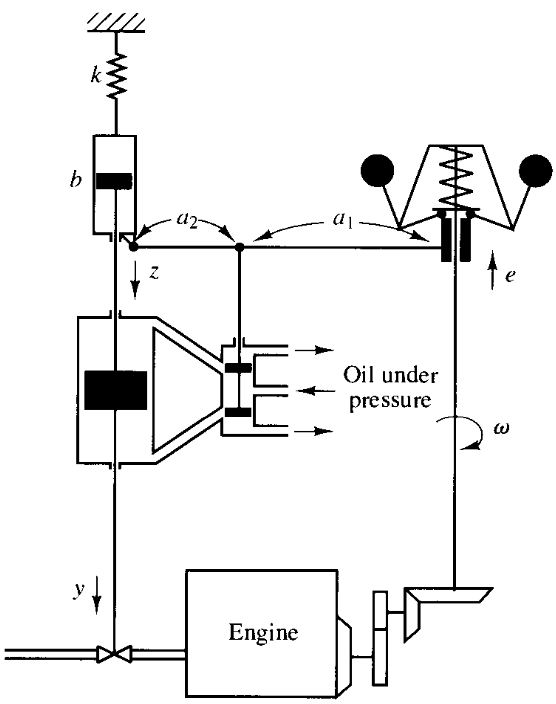


Figure 5–76
Speed control system.

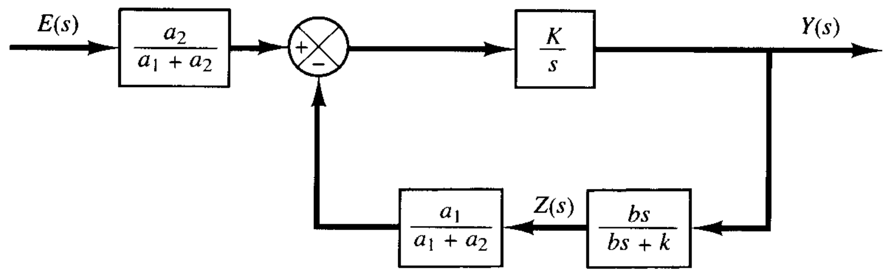


Figure 5–77 Block diagram for the speed control system shown in Figure 5–76.

If the following condition applies,

$$\left|\frac{a_1}{a_1 + a_2}\frac{bs}{bs + k}\frac{K}{s}\right| \gg 1$$

the transfer function $Y(s)/E(s)$ becomes

$$\frac{Y(s)}{E(s)} \doteqdot \frac{a_2}{a_1 + a_2}\frac{a_1 + a_2}{a_1}\frac{bs + k}{bs} = \frac{a_2}{a_1}\left(1 + \frac{k}{bs}\right)$$

The speed controller has proportional-plus-integral control action.

A-5-20. Consider the hydraulic servo system shown in Figure 5–78. Assuming that signal $e(t)$ is the input and power piston displacement $y(t)$ the output, find the transfer function $Y(s)/E(s)$.

Solution. A block diagram for the system can be drawn as shown in Figure 5–79. Assuming that $|K_1a_1/[s(a_1 + a_2)]| \gg 1$ and $|K_2b_1/[s(b_1 + b_2)]| \gg 1$, we obtain

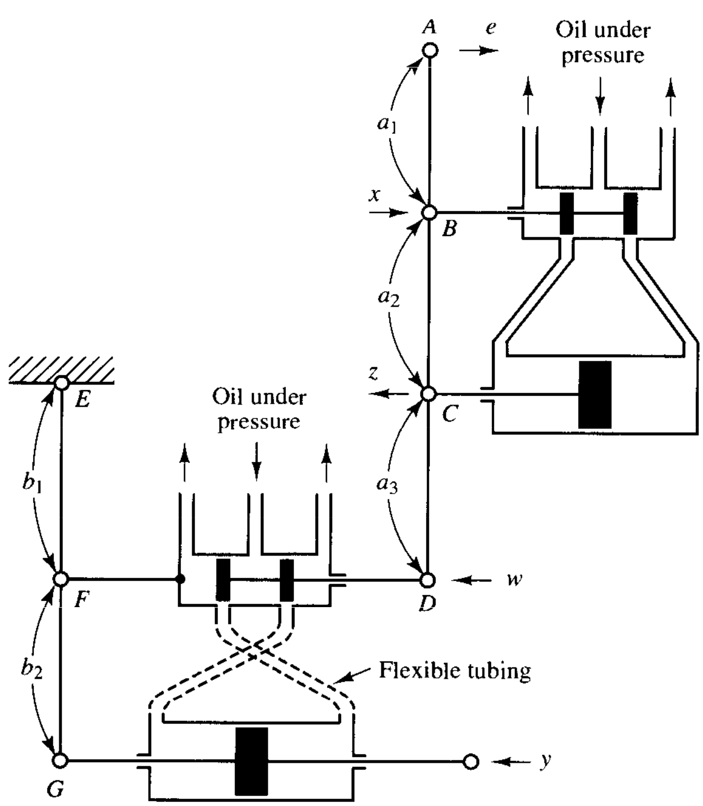


Figure 5–78 Hydraulic servo system.

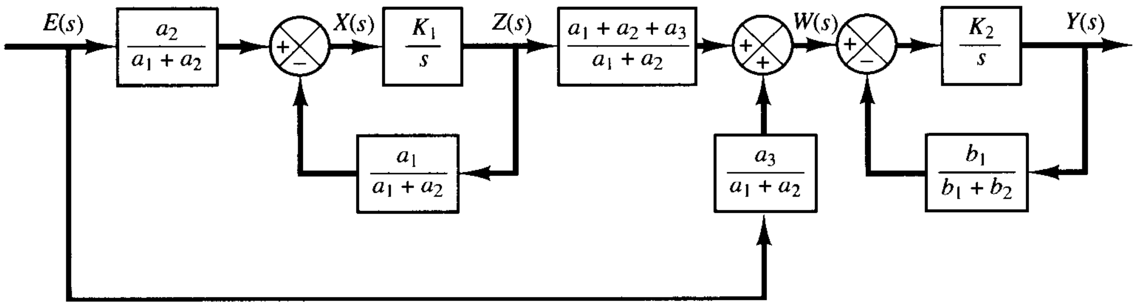


Figure 5–79
Block diagram for the system shown in Figure 5–78.

$$\frac{Z(s)}{E(s)} = \frac{\dfrac{a_2}{a_1 + a_2} \cdot \dfrac{K_1}{s}}{1 + \dfrac{K_1}{s} \cdot \dfrac{a_1}{a_1 + a_2}} \doteq \frac{a_2}{a_1 + a_2} \cdot \frac{a_1 + a_2}{a_1} = \frac{a_2}{a_1}$$

$$\frac{W(s)}{E(s)} = \frac{a_1 + a_2 + a_3}{a_1 + a_2} \cdot \frac{Z(s)}{E(s)} + \frac{a_3}{a_1 + a_2} = \frac{a_2 + a_3}{a_1}$$

$$\frac{Y(s)}{W(s)} = \frac{\dfrac{K_2}{s}}{1 + \dfrac{b_1}{b_1 + b_2} \dfrac{K_2}{s}} \doteq \frac{b_1 + b_2}{b_1}$$

Hence

$$\frac{Y(s)}{E(s)} = \frac{Y(s)}{W(s)} \cdot \frac{W(s)}{E(s)} = \frac{(a_2 + a_3)(b_1 + b_2)}{a_1 b_1}$$

This servo system is a proportional controller.

A-5-21. Obtain the transfer function $E_o(s)/E_i(s)$ of the op-amp circuit shown in Figure 5–80.

Solution. Define the voltage at point A as e_A. Then

$$\frac{E_A(s)}{E_i(s)} = \frac{R_1}{\dfrac{1}{Cs} + R_1} = \frac{R_1 Cs}{R_1 Cs + 1}$$

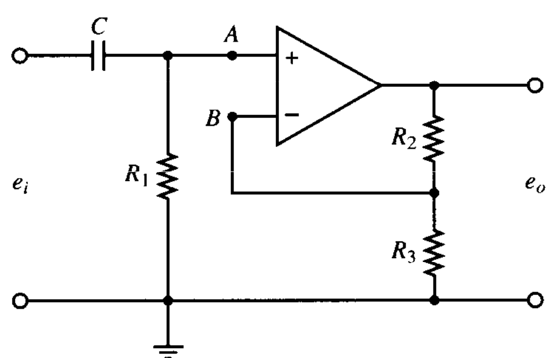


Figure 5–80
Operational-amplifier circuit.

Define the voltage at point B as e_B. Then

$$E_B(s) = \frac{R_3}{R_2 + R_3} E_o(s)$$

Noting that

$$[E_A(s) - E_B(s)]K = E_o(s)$$

and $K \gg 1$, we must have

$$E_A(s) = E_B(s)$$

Hence

$$E_A(s) = \frac{R_1Cs}{R_1Cs + 1} E_i(s) = E_B(s) = \frac{R_3}{R_2 + R_3} E_o(s)$$

from which we obtain

$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 + R_3}{R_3} \frac{R_1Cs}{R_1Cs + 1} = \frac{\left(1 + \frac{R_2}{R_3}\right)s}{s + \frac{1}{R_1C}}$$

A-5-22. Obtain the transfer function $E_o(s)/E_i(s)$ of the op-amp circuit shown in Figure 5–81.

Solution. The voltage at point A is

$$e_A = \frac{1}{2}(e_i - e_o) + e_o$$

The Laplace-transformed version of this last equation is

$$E_A(s) = \frac{1}{2}[E_i(s) + E_o(s)]$$

The voltage at point B is

$$E_B(s) = \frac{\frac{1}{Cs}}{R_2 + \frac{1}{Cs}} E_i(s) = \frac{1}{R_2Cs + 1} E_i(s)$$

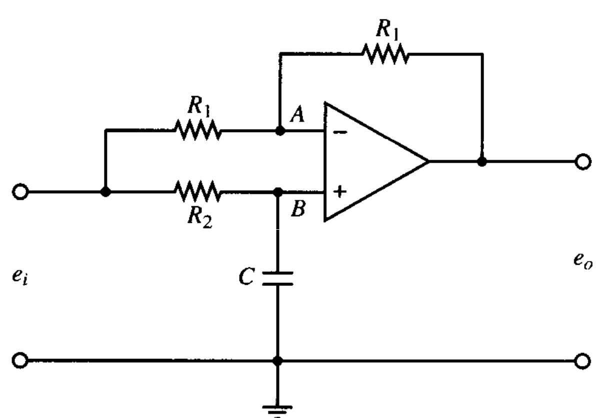


Figure 5–81
Operational-amplifier circuit.

Since $[E_B(s) - E_A(s)]K = E_o(s)$ and $K \gg 1$, we must have $E_A(s) = E_B(s)$. Thus

$$\frac{1}{2}[E_i(s) + E_o(s)] = \frac{1}{R_2Cs + 1}E_i(s)$$

Hence

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2Cs - 1}{R_2Cs + 1} = -\frac{s - \dfrac{1}{R_2C}}{s + \dfrac{1}{R_2C}}$$

A-5-23. Consider the stable unity-feedback control system with feedforward transfer function $G(s)$. Suppose that the closed-loop transfer function can be written

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{(T_as + 1)(T_bs + 1)\cdots(T_ms + 1)}{(T_1s + 1)(T_2s + 1)\cdots(T_ns + 1)} \qquad (m \le n)$$

Show that

$$\int_0^\infty e(t)\,dt = (T_1 + T_2 + \cdots + T_n) - (T_a + T_b + \cdots + T_m)$$

where $e(t)$ is the error in the unit-step response. Show also that

$$\frac{1}{\lim_{s\to 0} sG(s)} = (T_1 + T_2 + \cdots + T_n) - (T_a + T_b + \cdots + T_m)$$

Solution. Let us define

$$(T_as + 1)(T_bs + 1)\cdots(T_ms + 1) = P(s)$$

and

$$(T_1s + 1)(T_2s + 1)\cdots(T_ns + 1) = Q(s)$$

Then

$$\frac{C(s)}{R(s)} = \frac{P(s)}{Q(s)}$$

and

$$E(s) = \frac{Q(s) - P(s)}{Q(s)}R(s)$$

For a unit-step input, $R(s) = 1/s$ and

$$E(s) = \frac{Q(s) - P(s)}{sQ(s)}$$

Since the system is stable, $\int_0^\infty e(t)\,dt$ converges to a constant value. Referring to Table 2–2 (item 9), we have

$$\int_0^\infty e(t)\,dt = \lim_{s\to 0} s\frac{E(s)}{s} = \lim_{s\to 0} E(s)$$

Hence

$$\int_0^\infty e(t)\,dt = \lim_{s\to 0} \frac{Q(s) - P(s)}{sQ(s)}$$

$$= \lim_{s\to 0} \frac{Q'(s) - P'(s)}{Q(s) + sQ'(s)}$$

$$= \lim_{s\to 0} [Q'(s) - P'(s)]$$

Since

$$\lim_{s\to 0} P'(s) = T_a + T_b + \cdots + T_m$$

$$\lim_{s\to 0} Q'(s) = T_1 + T_2 + \cdots + T_n$$

we have

$$\int_0^\infty e(t)\,dt = (T_1 + T_2 + \cdots + T_n) - (T_a + T_b + \cdots + T_m)$$

For a unit-step input $r(t)$, since

$$\int_0^\infty e(t)\,dt = \lim_{s\to 0} E(s) = \lim_{s\to 0} \frac{1}{1 + G(s)} R(s) = \lim_{s\to 0} \frac{1}{1 + G(s)} \frac{1}{s} = \frac{1}{\lim\limits_{s\to 0} sG(s)} = \frac{1}{K_v}$$

we have

$$\frac{1}{\lim\limits_{s\to 0} sG(s)} = (T_1 + T_2 + \cdots + T_n) - (T_a + T_b + \cdots + T_m)$$

Note that zeros in the left half-plane (that is, positive $T_a, T_b, \ldots, T_m$) will improve K_v. Poles close to the origin cause low velocity-error constants unless there are zeros nearby.

PROBLEMS

B-5-1. If the feedforward path of a control system contains at least one integrating element, then the output continues to change as long as an error is present. The output stops when the error is precisely zero. If an external disturbance enters the system, it is desirable to have an integrating element between the error-measuring element and the point where the disturbance enters so that the effect of the external disturbance may be made zero at steady state.

Show that, if the disturbance is a ramp function, then the steady-state error due to this ramp disturbance may be eliminated only if two integrators precede the point where the disturbance enters.

B-5-2. Consider industrial automatic controllers whose control actions are proportional, integral, proportional-plus-integral, proportional-plus-derivative, and proportional-plus-integral-plus-derivative. The transfer functions of these controllers can be given, respectively, by

$$\frac{U(s)}{E(s)} = K_p$$

$$\frac{U(s)}{E(s)} = \frac{K_i}{s}$$

$$\frac{U(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s}\right)$$

$$\frac{U(s)}{E(s)} = K_p(1 + T_d s)$$

$$\frac{U(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

where $U(s)$ is the Laplace transform of $u(t)$, the controller output, and $E(s)$ the Laplace transform of $e(t)$, the actuating error signal. Sketch $u(t)$ versus t curves for each of the five types of controllers when the actuating error signal is

(a) $e(t)$ = unit-step function

(b) $e(t)$ = unit-ramp function

In sketching curves, assume that the numerical values of K_p, K_i, T_i, and T_d are given as

$$K_p = \text{proportional gain} = 4$$

$$K_i = \text{integral gain} = 2$$

$$T_i = \text{integral time} = 2 \text{ sec}$$

$$T_d = \text{derivative time} = 0.8 \text{ sec}$$

B-5-3. Consider a unity-feedback control system whose open-loop transfer function is

$$G(s) = \frac{K}{s(Js + B)}$$

Discuss the effects that varying the values of K and B has on the steady-state error in unit-ramp response. Sketch typical unit-ramp response curves for a small value, medium value, and large value of K.

B-5-4. Figure 5–82 shows three systems. System I is a positional servo system. System II is a positional servo system with PD control action. System III is a positional servo system with velocity feedback. Compare the unit-step, unit-impulse, and unit-ramp responses of the three systems. Which system is best with respect to the speed of response and maximum overshoot in the step response?

B-5-5. Consider the position control system shown in Figure 5–83. Write a MATLAB program to obtain a unit-step response and a unit-ramp response of the system. Plot curves $x_1(t)$ versus t, $x_2(t)$ versus t, $x_3(t)$ versus t, and $e(t)$ versus t [where $e(t) = r(t) - x_1(t)$] for both the unit-step response and the unit-ramp response.

B-5-6. Determine the range of K for stability of a unity-feedback control system whose open-loop transfer function is

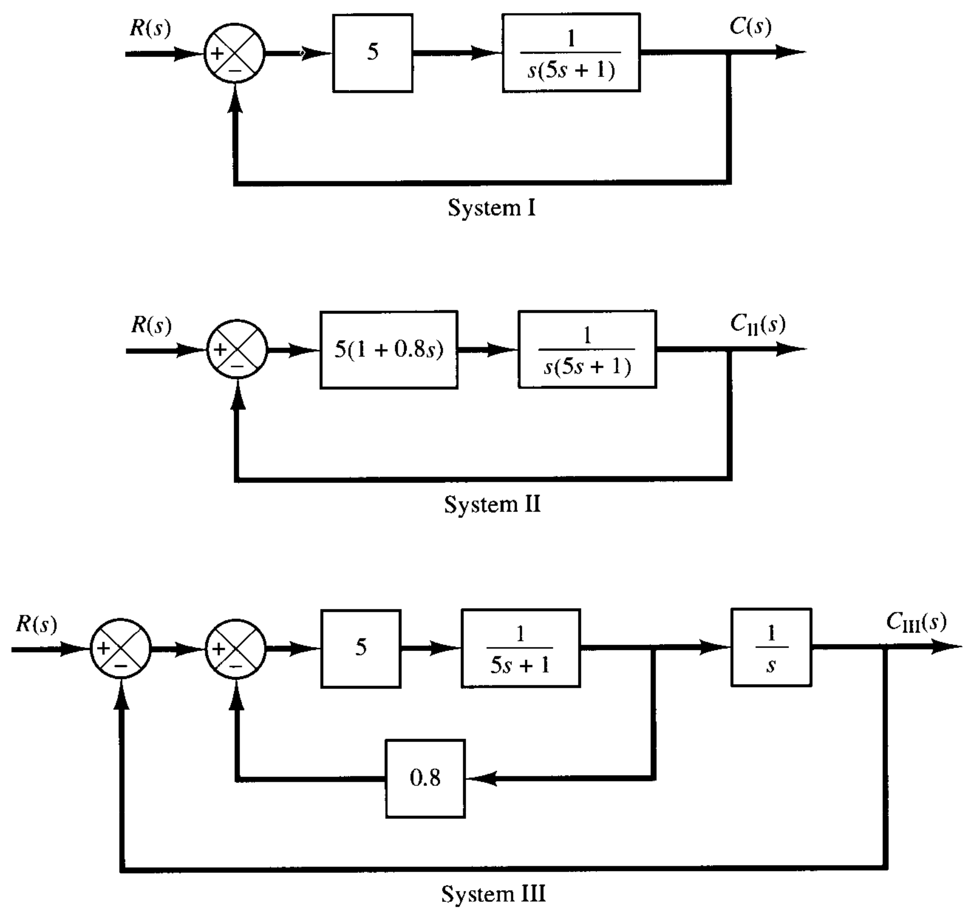


Figure 5–82
(a) Positional servo system; (b) positional servo system with PD control action; (c) positional servo system with velocity feedback.

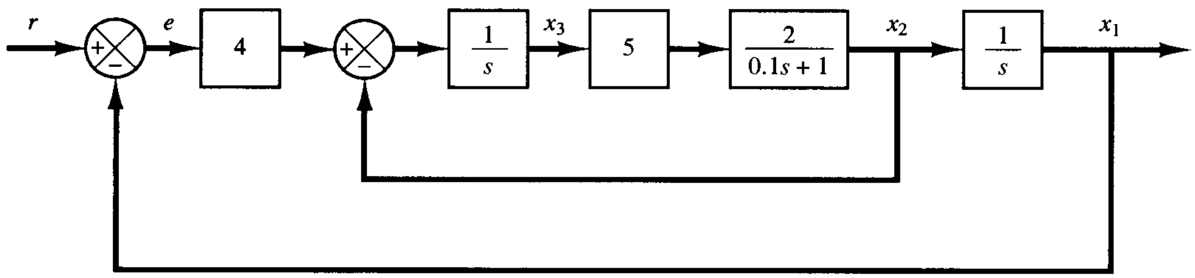


Figure 5–83
Position control system.

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

B-5-7. Consider the unity-feedback control system with the following open-loop transfer function:

$$G(s) = \frac{10}{s(s-1)(2s+3)}$$

Is this system stable?

B-5-8. Consider the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$$

where matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -b_3 & 0 & 1 \\ 0 & -b_2 & -b_1 \end{bmatrix}$$

($\mathbf{A}$ is called Schwarz matrix.) Show that the first column of the Routh's array of the characteristic equation $|s\mathbf{I} - \mathbf{A}| = 0$ consists of 1, b_1, b_2, and $b_1 b_3$.

B-5-9. Consider the pneumatic system shown in Figure 5–84. Obtain the transfer function $X(s)/P_i(s)$.

B-5-10. Figure 5–85 shows a pneumatic controller. What kind of control action does this controller produce? Derive the transfer function $P_c(s)/E(s)$.

B-5-11. Consider the pneumatic controller shown in Figure 5–86. Assuming that the pneumatic relay has the characteristics that $p_c = Kp_b$ (where $K > 0$), determine the control action of this controller. The input to the controller is e and the output is p_c.

B-5-12. Figure 5–87 shows a pneumatic controller. The signal e is the input and the change in the control pressure p_c is the output. Obtain the transfer function $P_c(s)/E(s)$. Assume that the pneumatic relay has the characteristics that $p_c = Kp_b$, where $K > 0$.

B-5-13. Consider the pneumatic controller shown in Figure 5–88. What control action does this controller produce? Assume that the pneumatic relay has the characteristics that $p_c = Kp_b$, where $K > 0$.

B-5-14. Figure 5–89 shows an electric-pneumatic transducer. Show that the change in the output pressure is proportional to the change in the input current.

B-5-15. Figure 5–90 shows a flapper valve. It is placed between two opposing nozzles. If the flapper is moved slightly to the right, the pressure unbalance occurs in the nozzles and the power piston moves to the left, and vice versa. Such a device is frequently used in hydraulic servos as the first-stage valve in two-stage servovalves. This usage occurs because considerable force may be needed to stroke larger spool valves that result from the steady-state flow force. To reduce or compensate this force, two-stage valve configuration is often employed; a flapper valve or jet pipe is used as the first-stage valve to provide a necessary force to stroke the second-stage spool valve.

Figure 5–91 shows a schematic diagram of a hydraulic servomotor in which the error signal is amplified in two stages using a jet pipe and a pilot valve. Draw a block diagram of the system of Figure 5–91 and then find the trans-

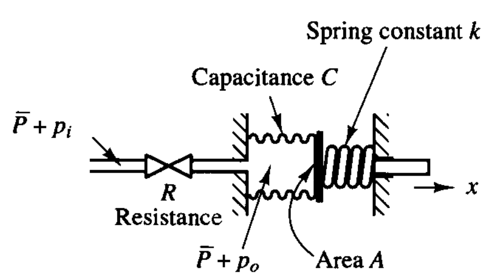


Figure 5–84
Pneumatic system.

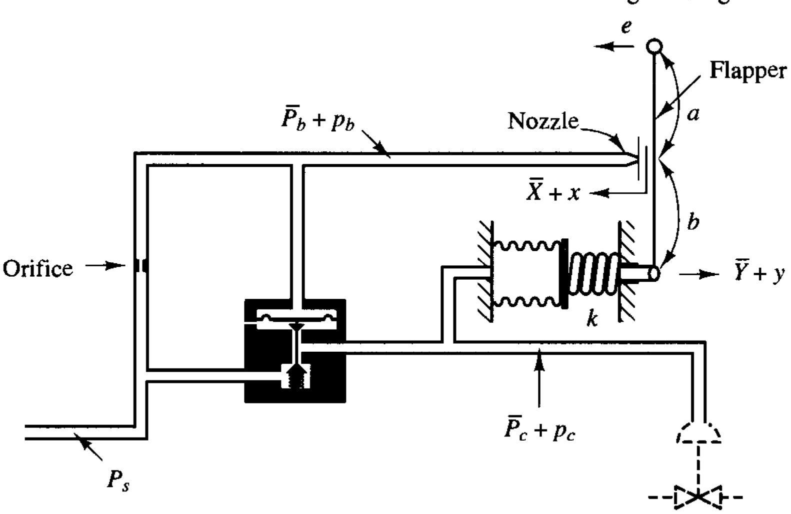


Figure 5–85 Pneumatic controller.

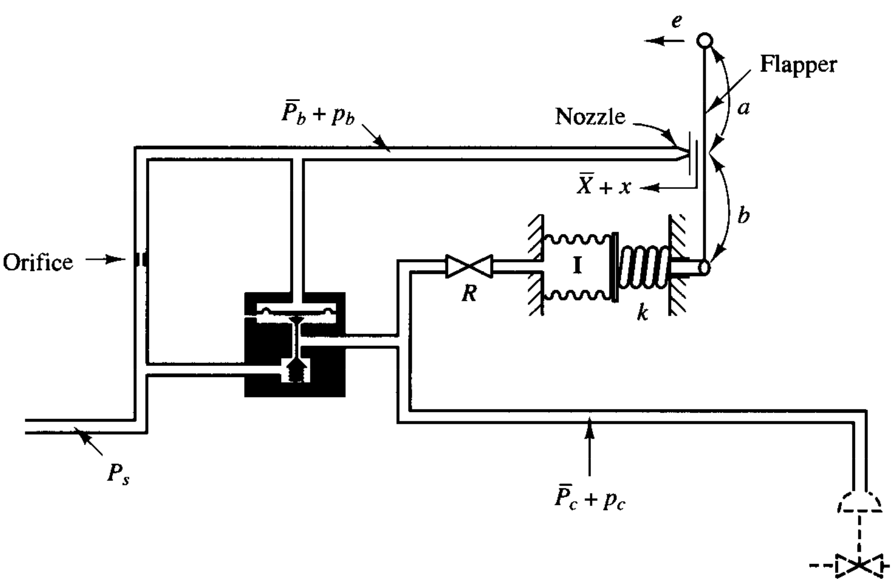


Figure 5–86 Pneumatic controller.

fer function between y and x, where x is the air pressure and y is the displacement of the power piston.

B-5-16. Figure 5–92 is a schematic diagram of an aircraft elevator control system. The input to the system is the deflection angle θ of the control lever, and the output is the elevator angle ϕ. Assume that angles θ and ϕ are relatively small. Show that for each angle θ of the control lever there is a corresponding (steady-state) elevator angle ϕ.

B-5-17. Consider the controller shown in Figure 5–93. The input is the air pressure p_i and the output is the displacement y of the power piston. Obtain the transfer function $Y(s)/P_i(s)$.

B-5-18. Obtain the transfer function $E_o(s)/E_i(s)$ of the op-amp circuit shown in Figure 5–94.

B-5-19. Obtain the transfer function $E_o(s)/E_i(s)$ of the op-amp circuit shown in Figure 5–95.

B-5-20. Consider a unity-feedback control system with the closed-loop transfer function

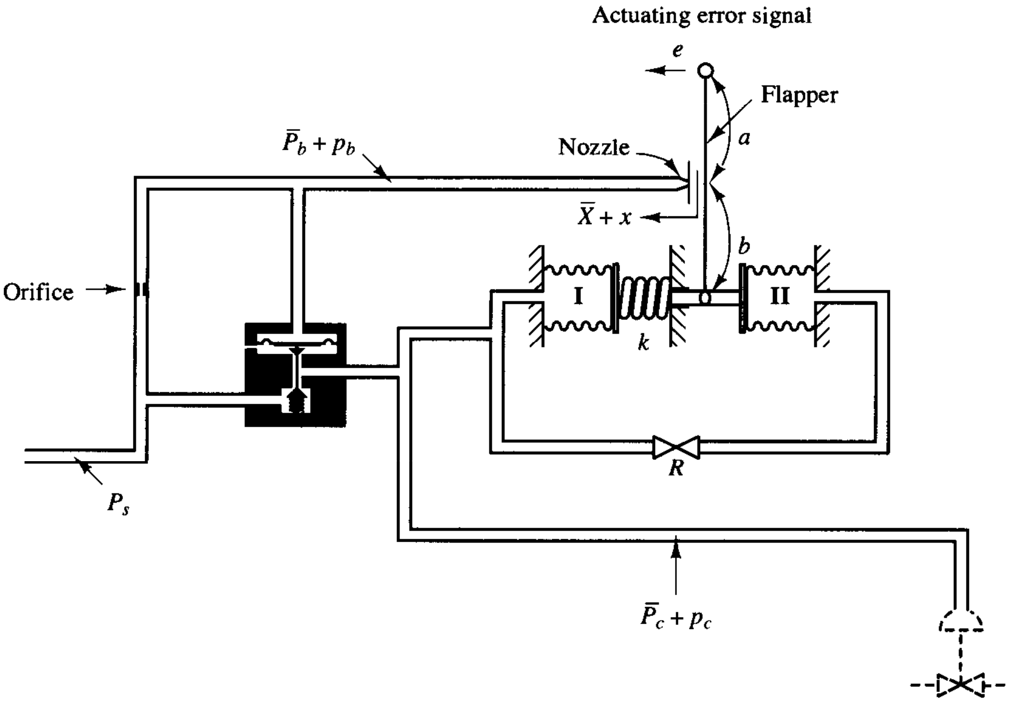


Figure 5–87 Pneumatic controller.

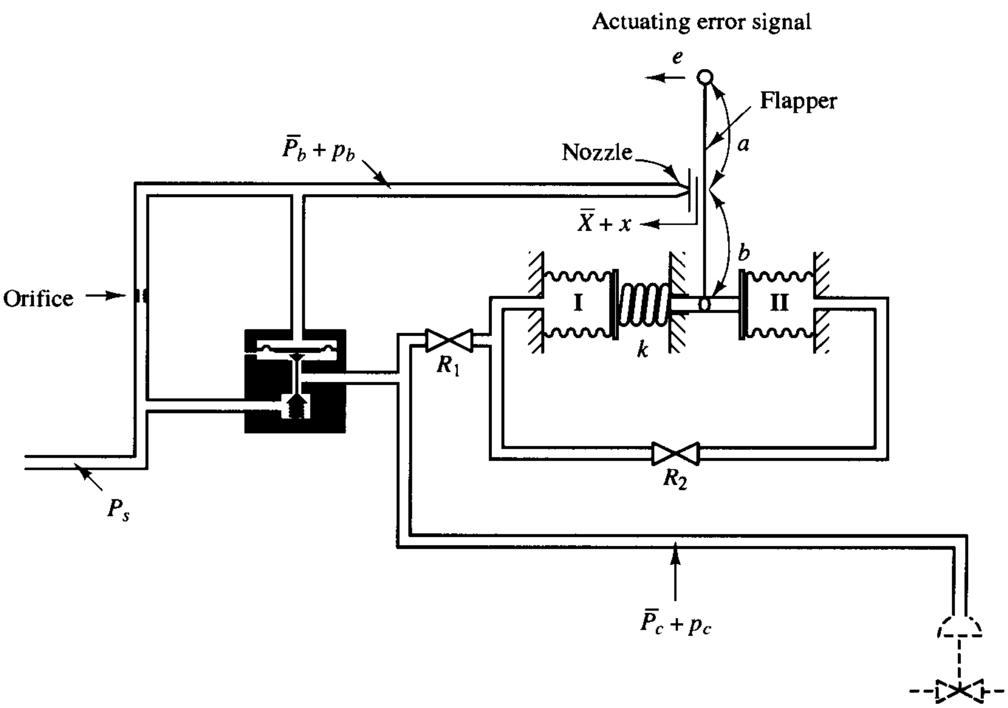


Figure 5–88 Pneumatic controller.

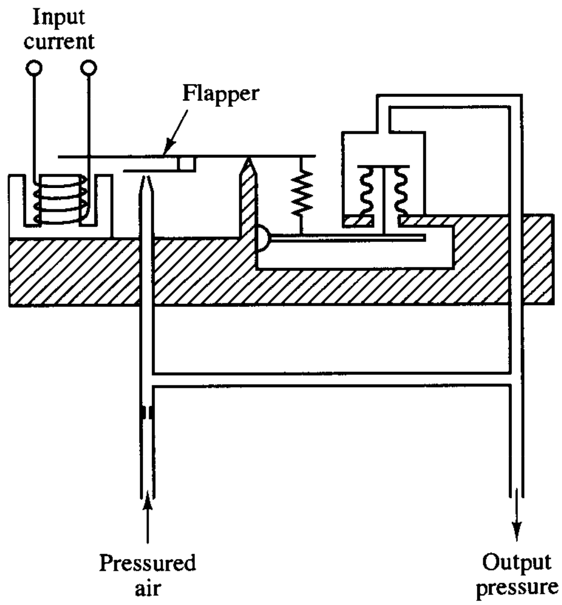


Figure 5–89
Electric-pneumatic transducer.

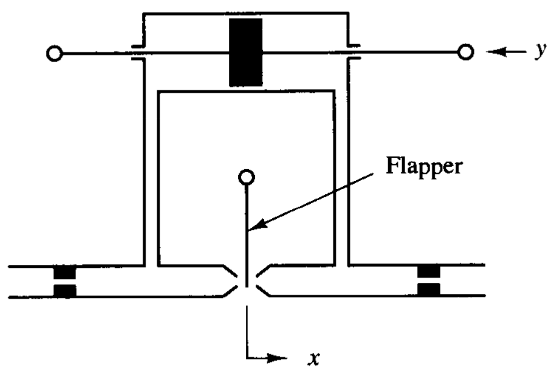


Figure 5–90
Flapper valve.

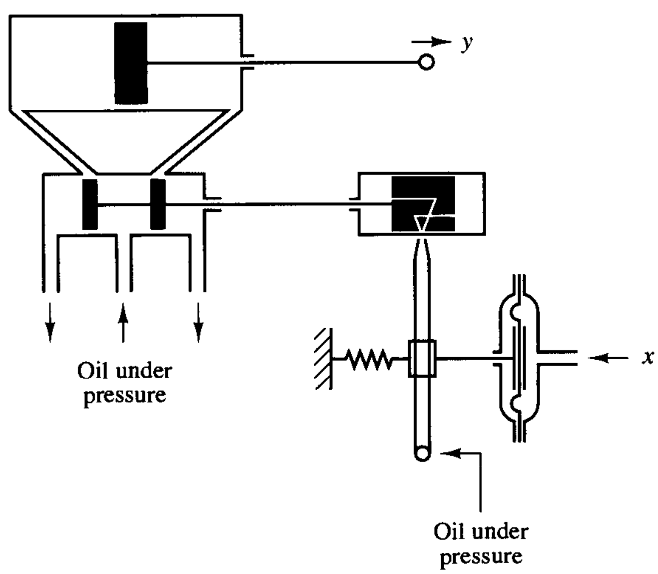


Figure 5–91
Schematic diagram of a hydraulic servomotor.

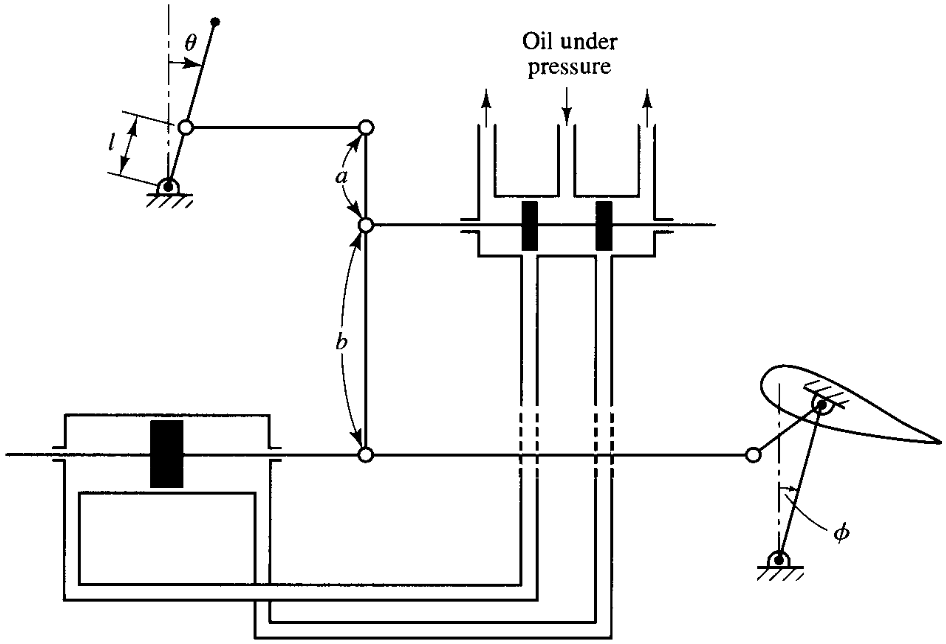


Figure 5–92
Aircraft elevator control system.

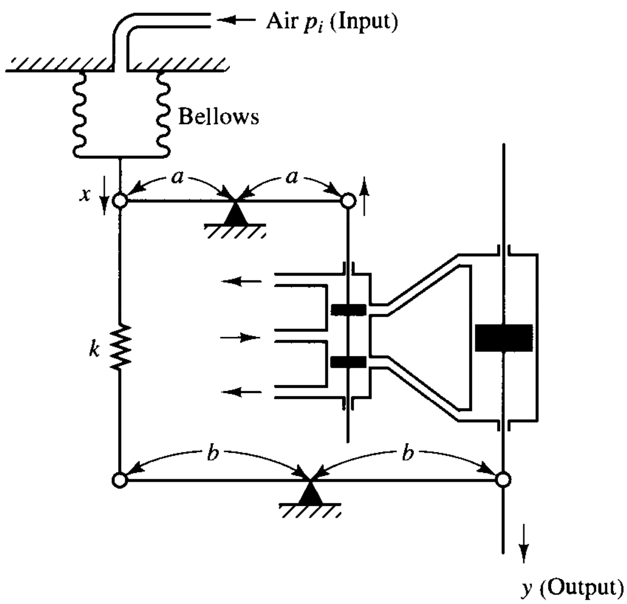


Figure 5–93
Controller.

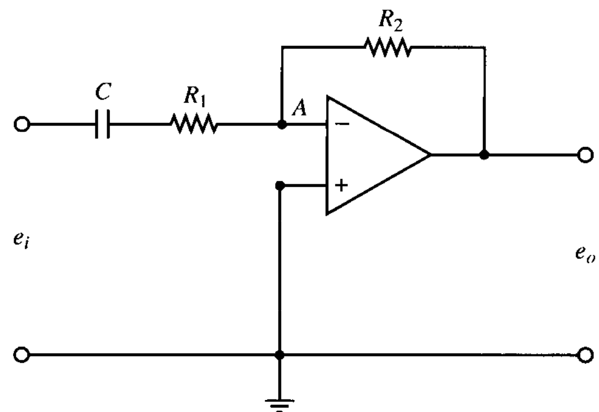


Figure 5–94
Operational-amplifier circuit.

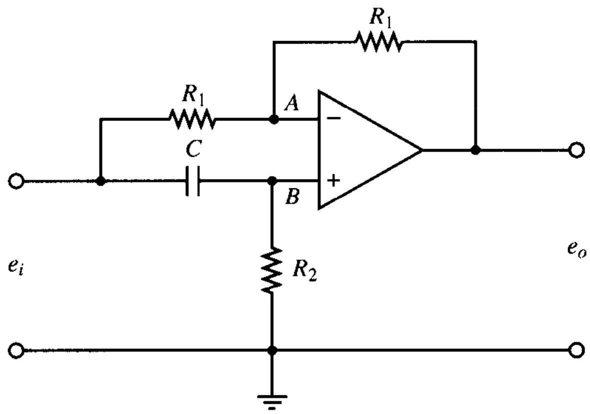


Figure 5–95
Operational amplifier circuit.

$$\frac{C(s)}{R(s)} = \frac{Ks + b}{s^2 + as + b}$$

Determine the open-loop transfer function $G(s)$.

Show that the steady-state error in the unit-ramp response is given by

$$e_{ss} = \frac{1}{K_v} = \frac{a - K}{b}$$

B-5-21. Show that the steady-state error in the response to ramp inputs can be made zero if the closed-loop transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{a_{n-1}s + a_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n}$$

6

Root-Locus Analysis

6-1 INTRODUCTION

The basic characteristic of the transient response of a closed-loop system is closely related to the location of the closed-loop poles. If the system has a variable loop gain, then the location of the closed-loop poles depends on the value of the loop gain chosen. It is important, therefore, that the designer know how the closed-loop poles move in the s plane as the loop gain is varied.

From the design viewpoint, in some systems simple gain adjustment may move the closed-loop poles to desired locations. Then the design problem may become the selection of an appropriate gain value. If the gain adjustment alone does not yield a desired result, addition of a compensator to the system will become necessary. (This subject is discussed in detail in Chapter 7.)

The closed-loop poles are the roots of the characteristic equation. Finding the roots of the characteristic equation of degree higher than 3 is laborious and will need computer solution. (MATLAB provides a simple solution to this problem.) However, just finding the roots of the characteristic equation may be of limited value, because as the gain of the open-loop transfer function varies the characteristic equation changes and the computations must be repeated.

A simple method for finding the roots of the characteristic equation has been developed by W. R. Evans and used extensively in control engineering. This method, called the *root-locus method,* is one in which the roots of the characteristic equation are plotted for all values of a system parameter. The roots corresponding to a particular value of this parameter can then be located on the resulting graph. Note that the parameter is usually the gain, but any other variable of the open-loop transfer function may be used. Unless otherwise stated, we shall assume that the gain of the open-loop transfer function is the parameter to be varied through all values, from zero to infinity.

By using the root-locus method the designer can predict the effects on the location of the closed-loop poles of varying the gain value or adding open-loop poles and/or open-loop zeros. Therefore, it is desired that the designer have a good understanding of the method for generating the root loci of the closed-loop system, both by hand and by use of a computer software like MATLAB.

Root-locus method. The basic idea behind the root-locus method is that the values of s that make the transfer function around the loop equal -1 must satisfy the characteristic equation of the system.

The locus of roots of the characteristic equation of the closed-loop system as the gain is varied from zero to infinity gives the method its name. Such a plot clearly shows the contributions of each open-loop pole or zero to the locations of the closed-loop poles.

In designing a linear control system, we find that the root-locus method proves quite useful since it indicates the manner in which the open-loop poles and zeros should be modified so that the response meets system performance specifications. This method is particularly suited to obtaining approximate results very quickly.

Some control systems may involve more than one parameter to be adjusted. The root-locus diagram for a system having multiple parameters may be constructed by varying one parameter at a time. In this chapter we include the discussion of the root loci for a system having two parameters. The root loci for such a case is called the *root contour.*

The root-locus method is a very powerful graphical technique for investigating the effects of the variation of a system parameter on the location of the closed-loop poles. In most cases, the system parameter is the loop gain K, although the parameter can be any other variable of the system. If the designer follows the general rules for constructing the root loci, sketching the root loci of a given system may become a simple matter.

Because generating the root loci by use of MATLAB is a very simple matter, one may think sketching the root loci by hand is a waste of time and effort. However, experience in sketching the root loci by hand is invaluable for interpreting computer-generated root loci, as well as for getting a rough idea of the root loci very quickly.

By using the root-locus method, it is possible to determine the value of the loop gain K that will make the damping ratio of the dominant closed-loop poles as prescribed. If the location of an open-loop pole or zero is a system variable, then the root-locus method suggests the way to choose the location of an open-loop pole or zero. (See Example 6–8 and Problems A-6-12 through A-6-14.) More on the control system design based on the root-locus method will be given in Chapter 7.

Outline of the chapter. This chapter introduces the basic concept of the root-locus method and presents useful rules for graphically constructing the root loci, as well as the generation of root loci with MATLAB.

The outline of the chapter is as follows: Section 6–1 has presented an introduction to the root-locus method. Section 6–2 details the concepts underlying the root-locus method and presents the general procedure for sketching root loci using illustrative examples. Section 6–3 summarizes general rules for constructing root loci, and Section 6–4 discusses generating root-locus plots with MATLAB. Section 6–5 treats special cases: the first case occurs when the variable K does not appear as a multiplicative factor and the second case when the closed-loop system has positive feedback. Section 6–6 analyzes closed-loop systems by use of the root-locus method. Section 6–7 extends the root-locus method to treat closed-loop systems with transport lag. Finally, Section 6–8 discusses root-contour plots.

6–2 ROOT-LOCUS PLOTS

Angle and magnitude conditions. Consider the system shown in Figure 6–1. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \tag{6–1}$$

The characteristic equation for this closed-loop system is obtained by setting the denominator of the right-hand side of Equation (6–1) equal to zero. That is,

$$1 + G(s)H(s) = 0$$

or

$$G(s)H(s) = -1 \tag{6–2}$$

Here we assume that $G(s)H(s)$ is a ratio of polynomials in s. [Later in Section 6–7 we extend the analysis to the case when $G(s)H(s)$ involves the transport lag e^{-Ts}.] Since $G(s)H(s)$ is a complex quantity, Equation (6–2) can be split into two equations by equating the angles and magnitudes of both sides, respectively, to obtain the following:

Angle condition:

$$\underline{/G(s)H(s)} = \pm 180°(2k + 1) \qquad (k = 0, 1, 2, \ldots) \tag{6–3}$$

Magnitude condition:

$$|G(s)H(s)| = 1 \tag{6–4}$$

The values of s that fulfill both the angle and magnitude conditions are the roots of the characteristic equation, or the closed-loop poles. A plot of the points in the complex plane satisfying the angle condition alone is the root locus. The roots of the characteristic equation (the closed-loop poles) corresponding to a given value of the gain can be determined from the magnitude condition. The details of applying the angle and magnitude conditions to obtain the closed-loop poles are presented later in this section.

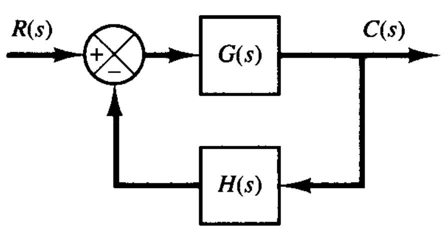


Figure 6–1
Control system.

In many cases, $G(s)H(s)$ involves a gain parameter K, and the characteristic equation may be written as

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0 \tag{6-5}$$

Then the root loci for the system are the loci of the closed-loop poles as the gain K is varied from zero to infinity.

Note that to begin sketching the root loci of a system by the root-locus method we must know the location of the poles and zeros of $G(s)H(s)$. Remember that the angles of the complex quantities originating from the open-loop poles and open-loop zeros to the test point s are measured in the counterclockwise direction. For example, if $G(s)H(s)$ is given by

$$G(s)H(s) = \frac{K(s + z_1)}{(s + p_1)(s + p_2)(s + p_3)(s + p_4)}$$

where $-p_2$ and $-p_3$ are complex-conjugate poles, then the angle of $G(s)H(s)$ is

$$\underline{/G(s)H(s)} = \phi_1 - \theta_1 - \theta_2 - \theta_3 - \theta_4$$

where ϕ_1, θ_1, θ_2, θ_3, and θ_4 are measured counterclockwise as shown in Figures 6–2(a) and (b). The magnitude of $G(s)H(s)$ for this system is

$$|G(s)H(s)| = \frac{KB_1}{A_1A_2A_3A_4}$$

where A_1, A_2, A_3, A_4, and B_1 are the magnitudes of the complex quantities $s + p_1, s + p_2, s + p_3, s + p_4$, and $s + z_1$, respectively, as shown in Figure 6–2(a).

Note that, because the open-loop complex-conjugate poles and complex-conjugate zeros, if any, are always located symmetrically about the real axis, the root loci are always

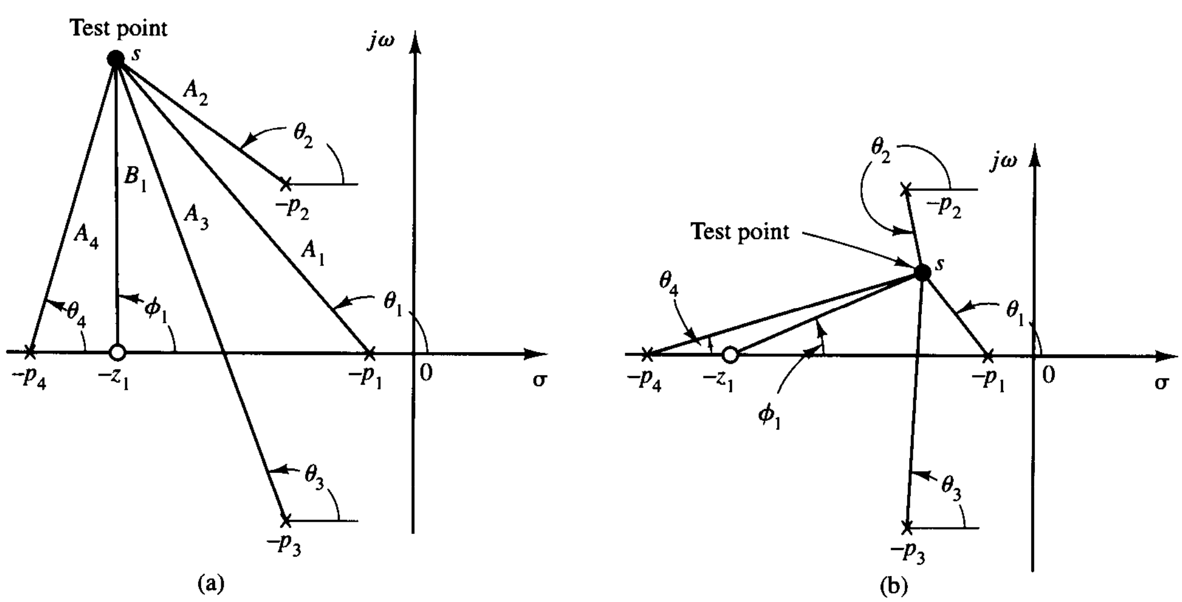


Figure 6–2
(a) and (b) Diagrams showing angle measurements from open-loop poles and open-loop zero to test point s.

symmetrical with respect to this axis. Therefore, we only need to construct the upper half of the root loci and draw the mirror image of the upper half in the lower-half s plane.

Illustrative examples. In what follows, two illustrative examples for constructing root-locus plots will be presented. Although computer approaches to the construction of the root loci are easily available, here we shall use graphical computation, combined with inspection, to determine the root loci upon which the roots of the characteristic equation of the closed-loop system must lie. Such a graphical approach will enhance understanding of how the closed-loop poles move in the complex plane as the open-loop poles and zeros are moved. Although we employ only simple systems for illustrative purposes, the procedure for finding the root loci is no more complicated for higher-order systems.

The first step in the procedure for constructing a root-locus plot is to seek out the loci of possible roots using the angle condition. Then, if necessary, the loci can be scaled, or graduated, in gain using the magnitude condition.

Because graphical measurements of angles and magnitudes are involved in the analysis, we find it necessary to use the same divisions on the abscissa as on the ordinate axis when sketching the root locus on graph paper.

EXAMPLE 6–1 Consider the system shown in Figure 6–3. (We assume that the value of gain K is nonnegative.) For this system,

$$G(s) = \frac{K}{s(s+1)(s+2)}, \qquad H(s) = 1$$

Let us sketch the root-locus plot and then determine the value of K such that the damping ratio ζ of a pair of dominant complex-conjugate closed-loop poles is 0.5.

For the given system, the angle condition becomes

$$\begin{aligned} \angle G(s) &= \angle \frac{K}{s(s+1)(s+2)} \\ &= -\angle s - \angle s+1 - \angle s+2 \\ &= \pm 180^\circ(2k+1) \qquad (k = 0, 1, 2, \ldots) \end{aligned}$$

The magnitude condition is

$$|G(s)| = \left| \frac{K}{s(s+1)(s+2)} \right| = 1$$

A typical procedure for sketching the root-locus plot is as follows:

1. *Determine the root loci on the real axis.* The first step in constructing a root-locus plot is to locate the open-loop poles, $s = 0$, $s = -1$, and $s = -2$, in the complex plane. (There are no

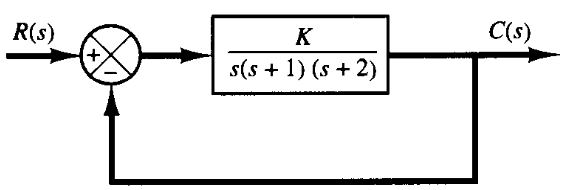


Figure 6–3
Control system.

open-loop zeros in this system.) The locations of the open-loop poles are indicated by crosses. (The locations of the open-loop zeros in this book will be indicated by small circles.) Note that the starting points of the root loci (the points corresponding to $K = 0$) are open-loop poles. The number of individual root loci for this system is three, which is the same as the number of open-loop poles.

To determine the root loci on the real axis, we select a test point, s. If the test point is on the positive real axis, then

$$\underline{/s} = \underline{/s+1} = \underline{/s+2} = 0°$$

This shows that the angle condition cannot be satisfied. Hence, there is no root locus on the positive real axis. Next, select a test point on the negative real axis between 0 and -1. Then

$$\underline{/s} = 180°, \qquad \underline{/s+1} = \underline{/s+2} = 0°$$

Thus

$$-\underline{/s} - \underline{/s+1} - \underline{/s+2} = -180°$$

and the angle condition is satisfied. Therefore, the portion of the negative real axis between 0 and -1 forms a portion of the root locus. If a test point is selected between -1 and -2, then

$$\underline{/s} = \underline{/s+1} = 180°, \qquad \underline{/s+2} = 0°$$

and

$$-\underline{/s} - \underline{/s+1} - \underline{/s+2} = -360°$$

It can be seen that the angle condition is not satisfied. Therefore, the negative real axis from -1 to -2 is not a part of the root locus. Similarly, if a test point is located on the negative real axis from -2 to $-\infty$, the angle condition is satisfied. Thus, root loci exist on the negative real axis between 0 and -1 and between -2 and $-\infty$.

2. *Determine the asymptotes of the root loci.* The asymptotes of the root loci as s approaches infinity can be determined as follows: If a test point s is selected very far from the origin, then

$$\lim_{s\to\infty} G(s) = \lim_{s\to\infty} \frac{K}{s(s+1)(s+2)} = \lim_{s\to\infty} \frac{K}{s^3}$$

and the angle condition becomes

$$-3\underline{/s} = \pm 180°(2k+1) \qquad (k = 0, 1, 2, \ldots)$$

or

$$\text{Angles of asymptotes} = \frac{\pm 180°(2k+1)}{3} \qquad (k = 0, 1, 2, \ldots)$$

Since the angle repeats itself as k is varied, the distinct angles for the asymptotes are determined as 60°, $-60°$, and 180°. Thus, there are three asymptotes. The one having the angle of 180° is the negative real axis.

Before we can draw these asymptotes in the complex plane, we must find the point where they intersect the real axis. Since

$$G(s) = \frac{K}{s(s+1)(s+2)} \tag{6-6}$$

if a test point is located very far from the origin, then $G(s)$ may be written as

$$G(s) = \frac{K}{s^3 + 3s^2 + \cdots} \tag{6–7}$$

Since the characteristic equation is

$$G(s) = -1$$

referring to Equation (6–7) the characteristic equation may be written as

$$s^3 + 3s^2 + \cdots = -K$$

For a large value of s, this last equation may be approximated by

$$(s + 1)^3 = 0$$

If the abscissa of the intersection of the asymptotes and the real axis is denoted by $s = -\sigma_a$, then

$$\sigma_a = -1$$

and the point of origin of the asymptotes is $(-1, 0)$. The asymptotes are almost part of the root loci in regions very far from the origin.

3. *Determine the breakaway point.* To plot root loci accurately, we must find the breakaway point, where the root-locus branches originating from the poles at 0 and -1 break away (as K is increased) from the real axis and move into the complex plane. The breakaway point corresponds to a point in the s plane where multiple roots of the characteristic equation occur.

A simple method for finding the breakaway point is available. We shall present this method in the following: Let us write the characteristic equation as

$$f(s) = B(s) + KA(s) = 0 \tag{6–8}$$

where $A(s)$ and $B(s)$ do not contain K. Note that $f(s) = 0$ has multiple roots at points where

$$\frac{df(s)}{ds} = 0$$

This can be seen as follows: Suppose that $f(s)$ has multiple roots of order r. Then $f(s)$ may be written as

$$f(s) = (s - s_1)^r(s - s_2) \cdots (s - s_n)$$

If we differentiate this equation with respect to s and set $s = s_1$, then we get

$$\left.\frac{df(s)}{ds}\right|_{s=s_1} = 0 \tag{6–9}$$

This means that multiple roots of $f(s)$ will satisfy Equation (6–9). From Equation (6–8), we obtain

$$\frac{df(s)}{ds} = B'(s) + KA'(s) = 0 \tag{6–10}$$

where

$$A'(s) = \frac{dA(s)}{ds}, \qquad B'(s) = \frac{dB(s)}{ds}$$

The particular value of K that will yield multiple roots of the characteristic equation is obtained from Equation (6–10) as

$$K = -\frac{B'(s)}{A'(s)}$$

If we substitute this value of K into Equation (6–8), we get

$$f(s) = B(s) - \frac{B'(s)}{A'(s)} A(s) = 0$$

or

$$B(s)A'(s) - B'(s)A(s) = 0 \tag{6–11}$$

If Equation (6–11) is solved for s, the points where multiple roots occur can be obtained. On the other hand, from Equation (6–8) we obtain

$$K = -\frac{B(s)}{A(s)}$$

and

$$\frac{dK}{ds} = -\frac{B'(s)A(s) - B(s)A'(s)}{A^2(s)}$$

If dK/ds is set equal to zero, we get the same equation as Equation (6–11). Therefore, the breakaway points can be simply determined from the roots of

$$\frac{dK}{ds} = 0$$

It should be noted that not all the solutions of Equation (6–11) or of $dK/ds = 0$ correspond to actual breakaway points. If a point at which $df(s)/ds = 0$ is on a root locus, it is an actual breakaway or break-in point. Stated differently, if at a point at which $df(s)/ds = 0$ the value of K takes a real positive value then that point is an actual breakaway or break-in point.

For the present example, the characteristic equation $G(s) + 1 = 0$ is given by

$$\frac{K}{s(s + 1)(s + 2)} + 1 = 0$$

or

$$K = -(s^3 + 3s^2 + 2s)$$

By setting $dK/ds = 0$, we obtain

$$\frac{dK}{ds} = -(3s^2 + 6s + 2) = 0$$

or

$$s = -0.4226, \qquad s = -1.5774$$

Since the breakaway point must lie on a root locus between 0 and -1, it is clear that $s = -0.4226$ corresponds to the actual breakaway point. Point $s = -1.5774$ is not on the root locus. Hence, this point is not an actual breakaway or break-in point. In fact, evaluation of the values of K corresponding to $s = -0.4226$ and $s = -1.5774$ yields

$$K = 0.3849, \qquad \text{for } s = -0.4226$$

$$K = -0.3849, \qquad \text{for } s = -1.5774$$

4. *Determine the points where the root loci cross the imaginary axis.* These points can be found by use of Routh's stability criterion as follows: Since the characteristic equation for the present system is

$$s^3 + 3s^2 + 2s + K = 0$$

the Routh array becomes

$$
\begin{array}{lcc}
s^3 & 1 & 2 \\
s^2 & 3 & K \\
s^1 & \dfrac{6-K}{3} & \\
s^0 & K &
\end{array}
$$

The value of K that makes the s^1 term in the first column equal zero is $K = 6$. The crossing points on the imaginary axis can then be found by solving the auxiliary equation obtained from the s^2 row; that is,

$$3s^2 + K = 3s^2 + 6 = 0$$

which yields

$$s = \pm j\sqrt{2}$$

The frequencies at the crossing points on the imaginary axis are thus $\omega = \pm\sqrt{2}$. The gain value corresponding to the crossing points is $K = 6$.

An alternative approach is to let $s = j\omega$ in the characteristic equation, equate both the real part and the imaginary part to zero, and then solve for ω and K. For the present system, the characteristic equation, with $s = j\omega$, is

$$(j\omega)^3 + 3(j\omega)^2 + 2(j\omega) + K = 0$$

or

$$(K - 3\omega^2) + j(2\omega - \omega^3) = 0$$

Equating both the real and imaginary parts of this last equation to zero, we obtain

$$K - 3\omega^2 = 0, \qquad 2\omega - \omega^3 = 0$$

from which

$$\omega = \pm\sqrt{2}, \qquad K = 6 \qquad \text{or} \qquad \omega = 0, \qquad K = 0$$

Thus, root loci cross the imaginary axis at $\omega = \pm\sqrt{2}$, and the value of K at the crossing points is 6. Also, a root-locus branch on the real axis touches the imaginary axis at $\omega = 0$.

5. *Choose a test point in the broad neighborhood of the $j\omega$ axis and the origin,* as shown in Figure 6–4, and apply the angle condition. If a test point is on the root loci, then the sum of the

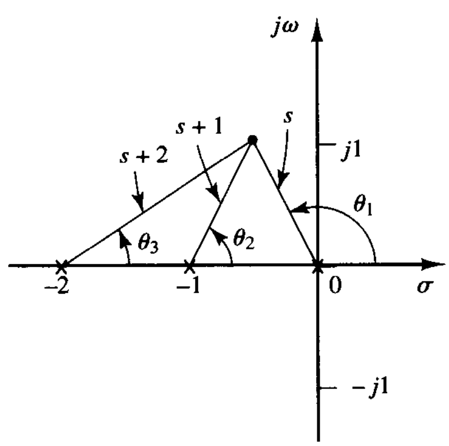


Figure 6–4
Construction of root locus.

three angles, $\theta_1 + \theta_2 + \theta_3$, must be 180°. If the test point does not satisfy the angle condition, select another test point until it satisfies the condition. (The sum of the angles at the test point will indicate which direction the test point should be moved.) Continue this process and locate a sufficient number of points satisfying the angle condition.

6. *Draw the root loci,* based on the information obtained in the foregoing steps, as shown in Figure 6–5.

7. *Determine a pair of dominant complex-conjugate closed-loop poles such that the damping ratio ζ is 0.5.* Closed-loop poles with $\zeta = 0.5$ lie on lines passing through the origin and making the angles $\pm\cos^{-1}\zeta = \pm\cos^{-1} 0.5 = \pm 60°$ with the negative real axis. From Figure 6–5, such closed-loop poles having $\zeta = 0.5$ are obtained as follows:

$$s_1 = -0.3337 + j0.5780, \qquad s_2 = -0.3337 - j0.5780$$

The value of K that yields such poles is found from the magnitude condition as follows:

$$\begin{aligned} K &= \left|s(s+1)(s+2)\right|_{s=-0.3337+j0.5780} \\ &= 1.0383 \end{aligned}$$

Using this value of K, the third pole is found at $s = -2.3326$.

Note that, from step 4, it can be seen that for $K = 6$ the dominant closed-loop poles lie on the imaginary axis at $s = \pm j\sqrt{2}$. With this value of K, the system will exhibit sustained oscillations. For $K > 6$, the dominant closed-loop poles lie in the right-half s plane, resulting in an unstable system.

Finally, note that, if necessary, the root loci can be easily graduated in terms of K by use of the magnitude condition. We simply pick out a point on a root locus, measure the magnitudes of the three complex quantities s, $s + 1$, and $s + 2$, and multiply these magnitudes; the product is equal to the gain value K at that point, or

$$|s| \cdot |s+1| \cdot |s+2| = K$$

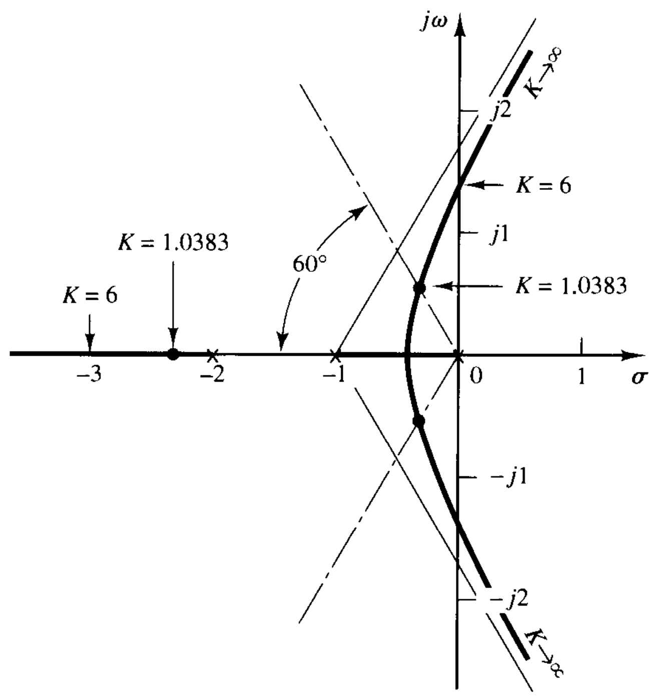


Figure 6–5
Root-locus plot.

EXAMPLE 6–2 In this example, we shall sketch the root-locus plot of a system with complex-conjugate open-loop poles. Consider the system shown in Figure 6–6. For this system,

$$G(s) = \frac{K(s+2)}{s^2+2s+3}, \qquad H(s) = 1$$

It is seen that $G(s)$ has a pair of complex-conjugate poles at

$$s = -1 + j\sqrt{2}, \qquad s = -1 - j\sqrt{2}$$

A typical procedure for sketching the root-locus plot is as follows:

1. *Determine the root loci on the real axis.* For any test point s on the real axis, the sum of the angular contributions of the complex-conjugate poles is 360°, as shown in Figure 6–7. Thus the net effect of the complex-conjugate poles is zero on the real axis. The location of the root locus on the real axis is determined from the open-loop zero on the negative real axis. A simple test reveals that a section of the negative real axis, that between -2 and $-\infty$, is a part of the root locus. It is noted that, since this locus lies between two zeros (at $s = -2$ and $s = -\infty$), it is actually a part of two root loci, each of which starts from one of the two complex-conjugate poles. In other words, two root loci break in the part of the negative real axis between -2 and $-\infty$.

Since there are two open-loop poles and one zero, there is one asymptote, which coincides with the negative real axis.

2. *Determine the angle of departure from the complex-conjugate open-loop poles.* The presence of a pair of complex-conjugate open-loop poles requires determination of the angle of departure from these poles. Knowledge of this angle is important since the root locus near a complex pole yields information as to whether the locus originating from the complex pole migrates toward the real axis or extends toward the asymptote.

Referring to Figure 6–8, if we choose a test point and move it in the very vicinity of the complex open-loop pole at $s = -p_1$, we find that the sum of the angular contributions from the pole at $s = -p_2$ and zero at $s = -z_1$ to the test point can be considered remaining the same. If the test

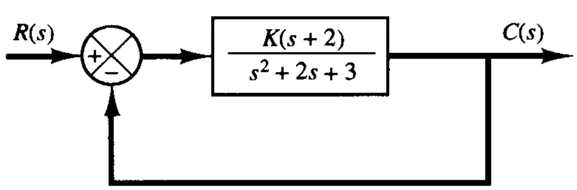


Figure 6–6
Control system.

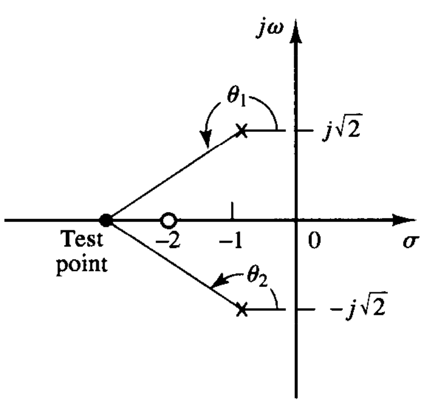


Figure 6–7
Determination of the root locus on the real axis.

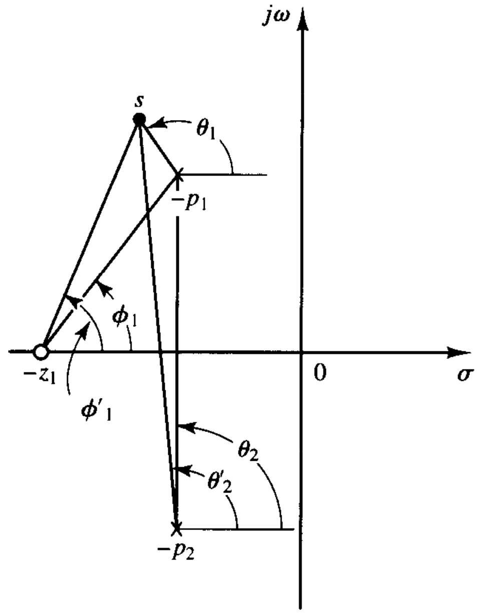


Figure 6–8
Determination of the angle of departure.

point is to be on the root locus, then the sum of ϕ_1', $-\theta_1$, and $-\theta_2'$ must be $\pm 180°(2k + 1)$, where $k = 0, 1, 2, \ldots$. Thus, in the example,

$$\phi_1' - (\theta_1 + \theta_2') = \pm 180°(2k + 1)$$

or

$$\theta_1 = 180° - \theta_2' + \phi_1' = 180° - \theta_2 + \phi_1$$

The angle of departure is then

$$\theta_1 = 180° - \theta_2 + \phi_1 = 180° - 90° + 55° = 145°$$

Since the root locus is symmetric about the real axis, the angle of departure from the pole at $s = -p_2$ is $-145°$.

3. *Determine the break-in point.* A break-in point exists where a pair of root-locus branches coalesce as K is increased. For this problem, the break-in point can be found as follows: Since

$$K = -\frac{s^2 + 2s + 3}{s + 2}$$

we have

$$\frac{dK}{ds} = -\frac{(2s + 2)(s + 2) - (s^2 + 2s + 3)}{(s + 2)^2} = 0$$

which gives

$$s^2 + 4s + 1 = 0$$

or

$$s = -3.7320 \qquad \text{or} \qquad s = -0.2680$$

Notice that point $s = -3.7320$ is on the root locus. Hence this point is an actual break-in point. (Note that at point $s = -3.7320$ the corresponding gain value is $K = 5.4641$.) Since point $s = -0.2680$ is not on the root locus, it cannot be a break-in point. (For point $s = -0.2680$, the corresponding gain value is $K = -1.4641$.)

4. *Sketch a root-locus plot,* based on the information obtained in the foregoing steps. To determine accurate root loci, several points must be found by trial and error between the break-in point and the complex open-loop poles. (To facilitate sketching the root-locus plot, we should find the direction in which the test point should be moved by mentally summing up the changes on the angles of the poles and zeros.) Figure 6–9 shows a complete root-locus plot for the system considered.

The value of the gain K at any point on root locus can be found by applying the magnitude condition. For example, the value of K at which the complex-conjugate closed-loop poles have the damping ratio $\zeta = 0.7$ can be found by locating the roots, as shown in Figure 6–9, and computing the value of K as follows:

$$K = \left|\frac{(s+1-j\sqrt{2})(s+1+j\sqrt{2})}{s+2}\right|_{s=-1.67+j1.70} = 1.34$$

It is noted that in this system the root locus in the complex plane is a part of a circle. Such a circular root locus will not occur in most systems. Circular root loci may occur in systems that involve two poles and one zero, two poles and two zeros, or one pole and two zeros. Even in such systems, whether circular root loci occur depends on the locations of poles and zeros involved.

To show the occurrence of a circular root locus in the present system, we need to derive the equation for the root locus. For the present system, the angle condition is

$$\underline{/s+2} - \underline{/s+1-j\sqrt{2}} - \underline{/s+1+j\sqrt{2}} = \pm 180°(2k+1)$$

If $s = \sigma + j\omega$ is substituted into this last equation, we obtain

$$\underline{/\sigma+2+j\omega} - \underline{/\sigma+1+j\omega-j\sqrt{2}} - \underline{/\sigma+1+j\omega+j\sqrt{2}} = \pm 180°(2k+1)$$

which can be written as

$$\tan^{-1}\left(\frac{\omega}{\sigma+2}\right) - \tan^{-1}\left(\frac{\omega-\sqrt{2}}{\sigma+1}\right) - \tan^{-1}\left(\frac{\omega+\sqrt{2}}{\sigma+1}\right) = \pm 180°(2k+1)$$

or

$$\tan^{-1}\left(\frac{\omega-\sqrt{2}}{\sigma+1}\right) + \tan^{-1}\left(\frac{\omega+\sqrt{2}}{\sigma+1}\right) = \tan^{-1}\left(\frac{\omega}{\sigma+2}\right) \pm 180°(2k+1)$$

Taking tangents of both sides of this last equation using the relationship

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y} \qquad (6\text{–}12)$$

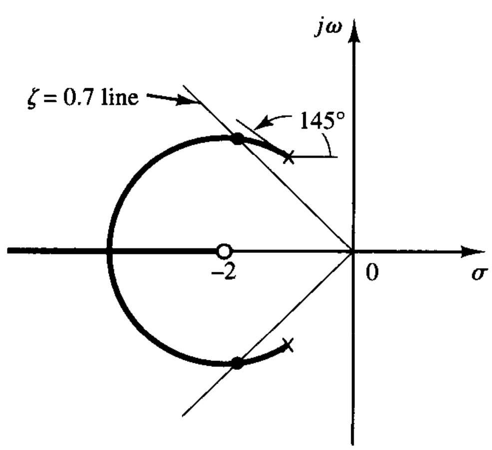


Figure 6–9
Root-locus plot.

we obtain

$$\tan\left[\tan^{-1}\left(\frac{\omega - \sqrt{2}}{\sigma + 1}\right) + \tan^{-1}\left(\frac{\omega + \sqrt{2}}{\sigma + 1}\right)\right] = \tan\left[\tan^{-1}\left(\frac{\omega}{\sigma + 2}\right) \pm 180^\circ(2k + 1)\right]$$

or

$$\frac{\dfrac{\omega - \sqrt{2}}{\sigma + 1} + \dfrac{\omega + \sqrt{2}}{\sigma + 1}}{1 - \left(\dfrac{\omega - \sqrt{2}}{\sigma + 1}\right)\left(\dfrac{\omega + \sqrt{2}}{\sigma + 1}\right)} = \frac{\dfrac{\omega}{\sigma + 2} \pm 0}{1 \mp \dfrac{\omega}{\sigma + 2} \times 0}$$

which can be simplified to

$$\frac{2\omega(\sigma + 1)}{(\sigma + 1)^2 - (\omega^2 - 2)} = \frac{\omega}{\sigma + 2}$$

or

$$\omega[(\sigma + 2)^2 + \omega^2 - 3] = 0$$

This last equation is equivalent to

$$\omega = 0 \qquad \text{or} \qquad (\sigma + 2)^2 + \omega^2 = (\sqrt{3})^2$$

These two equations are the equations for the root loci for the present system. Notice that the first equation, $\omega = 0$, is the equation for the real axis. The real axis from $s = -2$ to $s = -\infty$ corresponds to a root locus for $K \geq 0$. The remaining part of the real axis corresponds to a root locus when K is negative. (In the present system, K is nonnegative.) The second equation for the root locus is an equation of a circle with center at $\sigma = -2, \omega = 0$ and the radius equal to $\sqrt{3}$. That part of the circle to the left of the complex-conjugate poles corresponds to a root locus for $K \geq 0$. The remaining part of the circle corresponds to a root locus when K is negative.

It is important to note that easily interpretable equations for the root locus can be derived for simple systems only. For complicated systems having many poles and zeros, any attempt to derive equations for the root loci is discouraged. Such derived equations are very complicated and their configuration in the complex plane is difficult to visualize.

6–3 SUMMARY OF GENERAL RULES FOR CONSTRUCTING ROOT LOCI

For a complicated system with many open-loop poles and zeros, constructing a root-locus plot may seem complicated, but actually it is not difficult if the rules for constructing the root loci are applied. By locating particular points and asymptotes and by computing angles of departure from complex poles and angles of arrival at complex zeros, we can construct the general form of the root loci without difficulty.

Some of the rules for constructing root loci were given in Section 6–2. The purpose of this section is to summarize the general rules for constructing root loci of the system shown in Figure 6–10. While the root-locus method is essentially based on a trial-and-error technique, the number of trials required can be greatly reduced if we use these rules.

General rules for constructing root loci. We shall now summarize the general rules and procedure for constructing the root loci of the system shown in Figure 6–10.

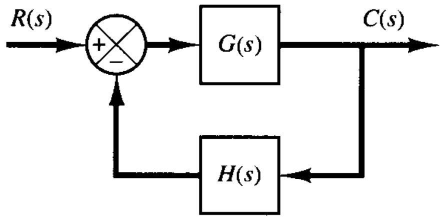


Figure 6–10
Control system.

First, obtain the characteristic equation

$$1 + G(s)H(s) = 0$$

Then rearrange this equation so that the parameter of interest appears as the multiplying factor in the form

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0 \tag{6–13}$$

In the present discussions, we assume that the parameter of interest is the gain K, where $K > 0$. (If $K < 0$, which corresponds to the positive-feedback case, the angle condition must be modified. See Section 6–5.) Note, however, that the method is still applicable to systems with parameters of interest other than gain.

1. *Locate the poles and zeros of $G(s)H(s)$ on the s plane. The root-locus branches start from open-loop poles and terminate at zeros (finite zeros or zeros at infinity).* From the factored form of the open-loop transfer function, locate the open-loop poles and zeros in the s plane. [Note that the open-loop zeros are the zeros of $G(s)H(s)$, while the closed-loop zeros consist of the zeros of $G(s)$ and the poles of $H(s)$.]

Note that the root loci are symmetrical about the real axis of the s plane, because the complex poles and complex zeros occur only in conjugate pairs.

Find the starting points and terminating points of the root loci and find also the number of separate root loci. The points on the root loci corresponding to $K = 0$ are open-loop poles. This can be seen from the magnitude condition by letting K approach zero, or

$$\lim_{K \to 0} \left| \frac{(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} \right| = \lim_{K \to 0} \frac{1}{K} = \infty$$

This last equation implies that as K is decreased the value of s must approach one of the open-loop poles. Each root locus thus originates at a pole of the open-loop transfer function $G(s)H(s)$. As K is increased to infinity, each root-locus approaches either a zero of the open-loop transfer function or infinity in the complex plane. This can be seen as follows: If we let K approach infinity in the magnitude condition, then

$$\lim_{K \to \infty} \left| \frac{(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} \right| = \lim_{K \to \infty} \frac{1}{K} = 0$$

Hence the value of s must approach one of the finite open-loop zeros or an open-loop zero at infinity. [If the zeros at infinity are included in the count, $G(s)H(s)$ has the same number of zeros as poles.]

A root-locus plot will have just as many branches as there are roots of the characteristic equation. Since the number of open-loop poles generally exceeds that of zeros, the number of branches equals that of poles. If the number of closed-loop poles is the same as the number of open-loop poles, then the number of individual root-locus branches terminating at finite open-loop zeros is equal to the number m of the open-loop zeros. The remaining $n - m$ branches terminate at infinity ($n - m$ implicit zeros at infinity) along asymptotes.

If we include poles and zeros at infinity, the number of open-loop poles is equal to that of open-loop zeros. Hence we can always state that the root loci start at the poles of $G(s)H(s)$ and end at the zeros of $G(s)H(s)$, as K increases from zero to infinity, where the poles and zeros include both those in the finite s plane and those at infinity.

2. *Determine the root loci on the real axis.* Root loci on the real axis are determined by open-loop poles and zeros lying on it. The complex-conjugate poles and zeros of the open-loop transfer function have no effect on the location of the root loci on the real axis because the angle contribution of a pair of complex-conjugate poles or zeros is 360° on the real axis. Each portion of the root locus on the real axis extends over a range from a pole or zero to another pole or zero. In constructing the root loci on the real axis, choose a test point on it. If the total number of real poles and real zeros to the right of this test point is odd, then this point lies on a root locus. The root locus and its complement form alternate segments along the real axis.

3. *Determine the asymptotes of root loci.* If the test point s is located far from the origin, then the angle of each complex quantity may be considered the same. One open-loop zero and one open-loop pole then cancel the effects of the other. Therefore, the root loci for very large values of s must be asymptotic to straight lines whose angles (slopes) are given by

$$\text{Angles of asymptotes} = \frac{\pm 180°(2k+1)}{n-m} \qquad (k = 0, 1, 2, \ldots)$$

where n = number of finite poles of $G(s)H(s)$
m = number of finite zeros of $G(s)H(s)$

Here, $k = 0$ corresponds to the asymptotes with the smallest angle with the real axis. Although k assumes an infinite number of values, as k is increased, the angle repeats itself, and the number of distinct asymptotes is $n - m$.

All the asymptotes intersect on the real axis. The point at which they do so is obtained as follows: If both the numerator and denominator of the open-loop transfer function are expanded, the result is

$$G(s)H(s) = \frac{K[s^m + (z_1 + z_2 + \cdots + z_m)s^{m-1} + \cdots + z_1 z_2 \cdots z_m]}{s^n + (p_1 + p_2 + \cdots + p_n)s^{n-1} + \cdots + p_1 p_2 \cdots p_n}$$

If a test point is located very far from the origin, then by dividing the denominator by the numerator $G(s)H(s)$ may be written as

$$G(s)H(s) = \frac{K}{s^{n-m} + [(p_1 + p_2 + \cdots + p_n) - (z_1 + z_2 + \cdots + z_m)]s^{n-m-1} + \cdots}$$

Since the characteristic equation is

$$G(s)H(s) = -1$$

it may be written

$$s^{n-m} + [(p_1 + p_2 + \cdots + p_n) - (z_1 + z_2 + \cdots + z_m)]s^{n-m-1} + \cdots = -K \tag{6–14}$$

For a large value of s, Equation (6–14) may be approximated by

$$\left[s + \frac{(p_1 + p_2 + \cdots + p_n) - (z_1 + z_2 + \cdots + z_m)}{n - m}\right]^{n-m} = 0$$

If the abscissa of the intersection of the asymptotes and the real axis is denoted by $s = \sigma_a$, then

$$\sigma_a = -\frac{(p_1 + p_2 + \cdots + p_n) - (z_1 + z_2 + \cdots + z_m)}{n - m} \tag{6–15}$$

or

$$\sigma_a = \frac{(\text{sum of poles}) - (\text{sum of zeros})}{n - m} \tag{6–16}$$

Because all the complex poles and zeros occur in conjugate pairs, σ_a is always a real quantity. Once the intersection of the asymptotes and the real axis is found, the asymptotes can be readily drawn in the complex plane.

It is important to note that the asymptotes show the behavior of the root loci for $|s| \gg 1$. A root locus branch may lie on one side of the corresponding asymptote or may cross the corresponding asymptote from one side to the other side.

4. *Find the breakaway and break-in points.* Because of the conjugate symmetry of the root loci, the breakaway points and break-in points either lie on the real axis or occur in complex-conjugate pairs.

If a root locus lies between two adjacent open-loop poles on the real axis, then there exists at least one breakaway point between the two poles. Similarly, if the root locus lies between two adjacent zeros (one zero may be located at $-\infty$) on the real axis, then there always exists at least one break-in point between the two zeros. If the root locus lies between an open-loop pole and a zero (finite or infinite) on the real axis, then there may exist no breakaway or break-in points or there may exist both breakaway and break-in points.

Suppose that the characteristic equation is given by

$$B(s) + KA(s) = 0$$

The breakaway points and break-in points correspond to multiple roots of the characteristic equation. Hence, the breakaway and break-in points can be determined from the roots of

$$\frac{dK}{ds} = -\frac{B'(s)A(s) - B(s)A'(s)}{A^2(s)} = 0 \tag{6–17}$$

where the prime indicates differentiation with respect to s. It is important to note that the breakaway points and break-in points must be the roots of Equation (6–17), but not all roots of Equation (6–17) are breakaway or break-in points. If a real root of Equation (6–17) lies on the root locus portion of the real axis, then it is an actual breakaway or break-in point. If a real root of Equation (6–17) is not on the root locus portion of the real axis, then this root corresponds to neither a breakaway point nor a break-in point. If two roots $s = s_1$ and $s = -s_1$ of Equation (6–17) are a complex-conjugate pair and if it is not certain whether they are on root loci, then it is necessary to check the corresponding K value. If the value of K corresponding to a root $s = s_1$ of $dK/ds = 0$ is positive, point $s = s_1$ is an actual breakaway or break-in point. (Since K is assumed to be nonnegative, if the value of K thus obtained is negative, then point $s = s_1$ is neither a breakaway nor break-in point.)

5. *Determine the angle of departure (angle of arrival) of the root locus from a complex pole (at a complex zero).* To sketch the root loci with reasonable accuracy, we must find the directions of the root loci near the complex poles and zeros. If a test point is chosen and moved in the very vicinity of a complex pole (or complex zero), the sum of the angular contributions from all other poles and zeros can be considered remaining the same. Therefore, the angle of departure (or angle of arrival) of the root locus from a complex pole (or at a complex zero) can be found by subtracting from 180° the sum of all the angles of vectors from all other poles and zeros to the complex pole (or complex zero) in question, with appropriate signs included.

Angle of departure from a complex pole = 180°
− (sum of the angles of vectors to a complex pole in question from other poles)
+ (sum of the angles of vectors to a complex pole in question from zeros)

Angle of arrival at a complex zero = 180°
− (sum of the angles of vectors to a complex zero in question from other zeros)
+ (sum of the angles of vectors to a complex zero in question from poles)

The angle of departure is shown in Figure 6–11.

6. *Find the points where the root loci may cross the imaginary axis.* The points where the root loci intersect the $j\omega$ axis can be found easily by (a) use of Routh's stability criterion or (b) letting $s = j\omega$ in the characteristic equation, equating both the real part and the imaginary part to zero, and solving for ω and K. The values of ω thus found give the frequencies at which root loci cross the imaginary axis. The K value corresponding to each crossing frequency gives the gain at the crossing point.

7. *Taking a series of test points in the broad neighborhood of the origin of the s plane, sketch the root loci.* Determine the root loci in the broad neighborhood of the $j\omega$ axis and the origin. The most important part of the root loci is on neither the real axis nor the asymptotes but the part in the broad neighborhood of the $j\omega$ axis and the origin. The

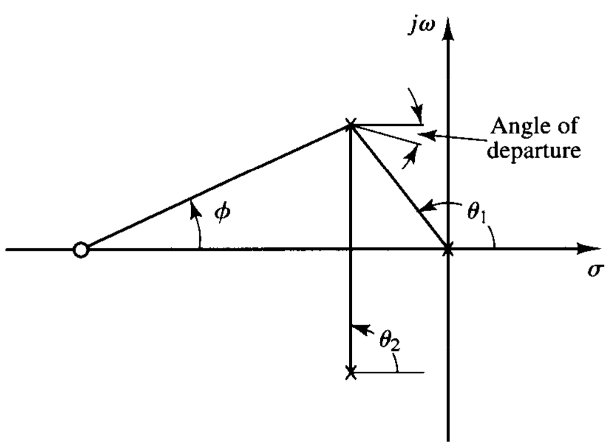


Figure 6–11
Construction of the root locus. [Angle of departure = 180° − $(\theta_1 + \theta_2) + \phi$.]

shape of the root loci in this important region in the s plane must be obtained with sufficient accuracy.

8. *Determine closed-loop poles.* A particular point on each root-locus branch will be a closed-loop pole if the value of K at that point satisfies the magnitude condition. Conversely, the magnitude condition enables us to determine the value of the gain K at any specific root location on the locus. (If necessary, the root loci may be graduated in terms of K. The root loci are continuous with K.)

The value of K corresponding to any point s on a root locus can be obtained using the magnitude condition, or

$$K = \frac{\text{product of lengths between point } s \text{ to poles}}{\text{product of lengths between point } s \text{ to zeros}}$$

This value can be evaluated either graphically or analytically.

If the gain K of the open-loop transfer function is given in the problem, then by applying the magnitude condition we can find the correct locations of the closed-loop poles for a given K on each branch of the root loci by a trial-and-error approach or by use of MATLAB, which will be presented in Section 6–4.

Comments on the root-locus plots. It is noted that the characteristic equation of the system whose open-loop transfer function is

$$G(s)H(s) = \frac{K(s^m + b_1 s^{m-1} + \cdots + b_m)}{s^n + a_1 s^{n-1} + \cdots + a_n} \qquad (n \geq m)$$

is an nth-degree algebraic equation in s. If the order of the numerator of $G(s)H(s)$ is lower than that of the denominator by two or more (which means that there are two or more zeros at infinity), then the coefficient a_1 is the negative sum of the roots of the equation and is independent of K. In such a case, if some of the roots move on the locus toward the left as K is increased, then the other roots must move toward the right as K is increased. This information is helpful in finding the general shape of the root loci.

It is also noted that a slight change in the pole–zero configuration may cause significant changes in the root-locus configurations. Figure 6–12 demonstrates the fact that a slight change in the location of a zero or pole will make the root-locus configuration look quite different.

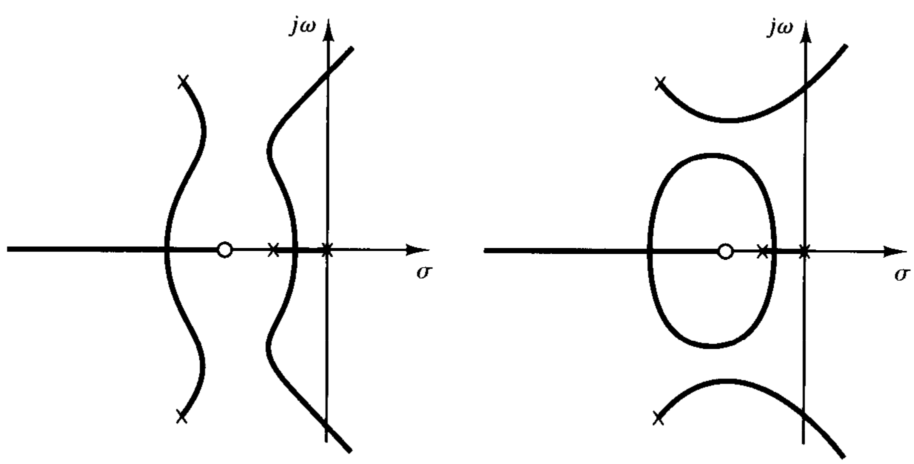


Figure 6–12
Root-locus plot.

Cancellation of poles $G(s)$ with zeros of $H(s)$. It is important to note that if the denominator of $G(s)$ and the numerator of $H(s)$ involve common factors then the corresponding open-loop poles and zeros will cancel each other, reducing the degree of the characteristic equation by one or more. For example, consider the system shown in Figure 6–13(a). (This system has velocity feedback.) By modifying the block diagram of Figure 6–13(a) to that shown in Figure 6–13(b), it is clearly seen that $G(s)$ and $H(s)$ have a common factor $s + 1$. The closed-loop transfer function $C(s)/R(s)$ is

$$\frac{C(s)}{R(s)} = \frac{K}{s(s+1)(s+2) + K(s+1)}$$

The characteristic equation is

$$[s(s+2) + K](s+1) = 0$$

Because of the cancellation of the terms $(s + 1)$ appearing in $G(s)$ and $H(s)$, however, we have

$$1 + G(s)H(s) = 1 + \frac{K(s+1)}{s(s+1)(s+2)}$$

$$= \frac{s(s+2) + K}{s(s+2)}$$

The reduced characteristic equation is

$$s(s+2) + K = 0$$

The root-locus plot of $G(s)H(s)$ does not show all the roots of the characteristic equation, only the roots of the reduced equation.

To obtain the complete set of closed-loop poles, we must add the canceled pole of $G(s)H(s)$ to those closed-loop poles obtained from the root-locus plot of $G(s)H(s)$. The

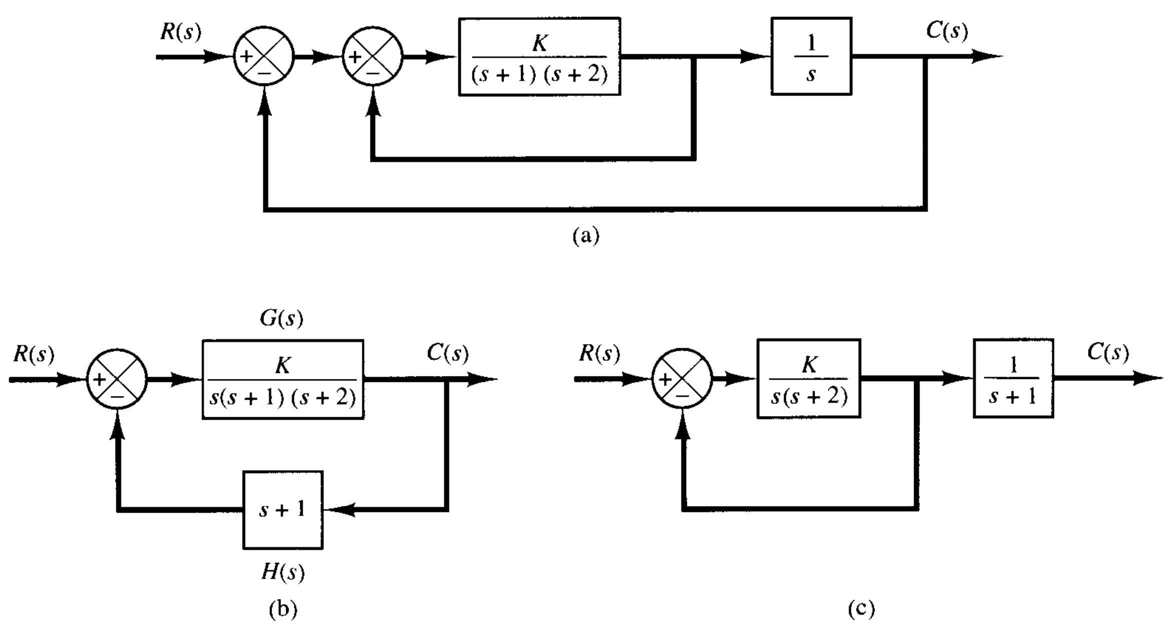
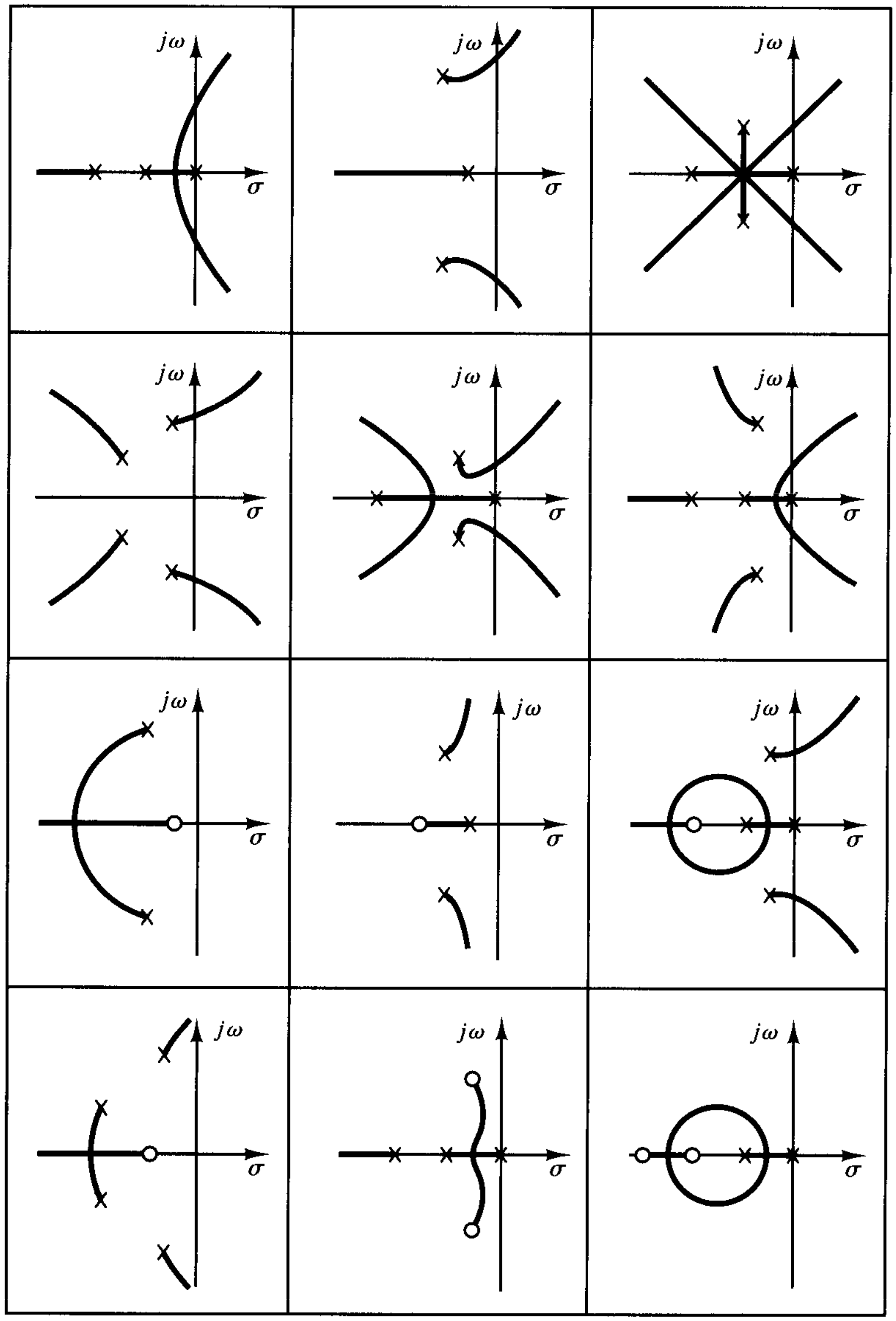


Figure 6–13
(a) Control system with velocity feedback; (b) and (c) modified block diagrams.

important thing to remember is that the canceled pole of $G(s)H(s)$ is a closed-loop pole of the system, as seen from Figure 6–13(c).

Typical pole–zero configurations and corresponding root loci. In concluding this section, we show several open-loop pole–zero configurations and their corresponding root loci in Table 6–1. The pattern of the root loci depends only on the relative separation of the open-loop poles and zeros. If the number of open-loop poles

Table 6–1 Open-Loop Pole–Zero Configurations and the Corresponding Root Loci



exceeds the number of finite zeros by three or more, there is a value of the gain K beyond which root loci enter the right-half s plane, and thus the system can become unstable. A stable system must have all its closed-loop poles in the left-half s plane.

Note that once we have some experience with the method we can easily evaluate the changes in the root loci due to the changes in the number and location of the open-loop poles and zeros by visualizing the root-locus plots resulting from various pole–zero configurations.

Summary. From the preceding discussions, it should be clear that it is possible to sketch a reasonably accurate root-locus diagram for a given system by following simple rules. (The reader is suggested to study various root-locus diagrams shown in the solved problems at the end of the chapter.) At preliminary design stages, we may not need the precise locations of the closed-loop poles. Often their approximate locations are all that is needed to make an estimate of system performance. Thus, it is important that the designer have the capability of quickly sketching the root loci for a given system.

6–4 ROOT-LOCUS PLOTS WITH MATLAB

In this section we present the MATLAB approach to the generation of root-locus plots.

Plotting root loci with MATLAB. In plotting root loci with MATLAB we deal with the system equation given in the form of Equation (6–13), which may be written as

$$1 + K\frac{\text{num}}{\text{den}} = 0$$

where num is the numerator polynomial and den is the denominator polynomial. That is,

$$\begin{aligned}
\text{num} &= (s + z_1)(s + z_2)\cdots(s + z_m) \\
&= s^m + (z_1 + z_2 + \cdots + z_m)s^{m-1} + \cdots + z_1 z_2 \cdots z_m \\
\text{den} &= (s + p_1)(s + p_2)\cdots(s + p_n) \\
&= s^n + (p_1 + p_2 + \cdots + p_n)s^{n-1} + \cdots + p_1 p_2 \cdots p_n
\end{aligned}$$

Note that both vectors num and den must be written in descending powers of s.

A MATLAB command commonly used for plotting root loci is

```
rlocus(num,den)
```

Using this command, the root-locus plot is drawn on the screen. The gain vector K is automatically determined. The command rlocus works for both continuous- and discrete-time systems.

For the systems defined in state space, rlocus(A,B,C,D) plots the root locus of the system with the gain vector automatically determined.

Note that commands

```
rlocus(num,den,K)        and        rlocus(A,B,C,D,K)
```

use the user-supplied gain vector K. (The vector K contains all the gain values for which the closed-loop poles are to be computed.)

If invoked with left-hand arguments

```
[r,K] = rlocus(num,den)
[r,K] = rlocus(num,den,K)
[r,K] = rlocus(A,B,C,D)
[r,K] = rlocus(A,B,C,D,K)
```

the screen will show the matrix r and gain vector K. (r has length K rows and length den − 1 columns containing the complex root locations. Each row of the matrix corresponds to a gain from vector K.) The plot command

```
plot(r,' ')
```

plots the root loci.

If it is desired to plot the root loci with marks 'o' or 'x', it is necessary to use the following command:

```
r = rlocus(num,den)
plot(r,'o')        or        plot(r,'x')
```

Plotting root loci using marks 'o' or 'x' is instructive, since each calculated closed-loop pole is graphically shown; in some portion of the root loci those marks are densely placed and in another portion of the root loci they are sparsely placed. MATLAB supplies its own set of gain values used to calculate a root-locus plot. It does so by an internal adaptive step-size routine. Also, MATLAB uses the automatic axis-scaling feature of the *plot* command.

Finally, note that, since the gain vector is automatically determined, root-locus plots of

$$G(s)H(s) = \frac{K(s+1)}{s(s+2)(s+3)}$$

$$G(s)H(s) = \frac{10K(s+1)}{s(s+2)(s+3)}$$

$$G(s)H(s) = \frac{200K(s+1)}{s(s+2)(s+3)}$$

are all the same. The num and den set of the system is the same for all three systems. The num and den are

```
num = [0  0  1  1]
den = [1  5  6  0]
```

EXAMPLE 6–3 Consider the control system shown in Figure 6–14. To plot the root-locus diagram with MATLAB, it is necessary to find the numerator and denominator polynomials of the open loop.

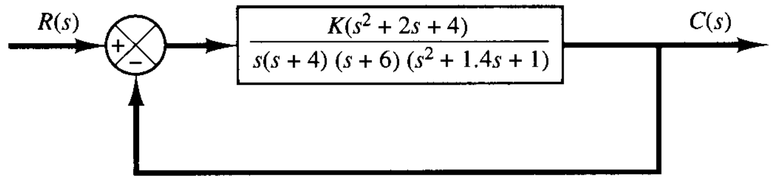


Figure 6–14
Control system.

For this problem, the numerator is already given as a polynomial in s. However, the denominator is given as a product of first- and second-order terms, with the result that we must multiply these terms to get a polynomial in s. The multiplication of these terms can be done easily by use of the *convolution command,* as shown next.

Define

$$a = s(s + 4) = s^2 + 4s \quad : \quad a = [1 \quad 4 \quad 0]$$
$$b = s + 6 \quad : \quad b = [1 \quad 6]$$
$$c = s^2 + 1.4s + 1 \quad : \quad c = [1 \quad 1.4 \quad 1]$$

Then use the following command:

```
d = conv(a,b);      e = conv(c,d)
```

[Note that conv(a,b) gives the product of two polynomials a and b.] See the following computer output:

```
a = [1   4   0];
b = [1   6];
c = [1   1.4   1];
d = conv(a,b)

d =

    1   10   24   0

e = conv(c,d)

e =

    1.0000   11.4000   39.0000   43.6000   24.0000   0
```

The denominator polynomial is thus found to be

```
den = [1   11.4   39   43.6   24   0]
```

To find the open-loop zeros of the given transfer function, we may use the following roots command:

```
p = [1   2   4]
r = roots(p)
```

The command and the computer output are shown next.

```
p = [1   2   4];
r = roots(p)

r =

  -1.0000 + 1.7321i
  -1.0000 - 1.7321i
```

Similarly, to find the complex-conjugate open-loop poles (the roots of $s^2 + 1.4s + 1 = 0$), we may enter the roots commands as follows:

```
q = roots(c)

q =

    -0.7000 + 0.7141i
    -0.7000 - 0.7141i
```

Thus the system has the following open-loop zeros and open-loop poles;

Open-loop zeros:	$s = -1 + j1.7321,$		$s = -1 - j1.7321$
Open-loop poles:	$s = -0.7 + j0.7141,$		$s = -0.7 - j0.7141$
	$s = 0,$	$s = -4,$	$s = -6$

MATLAB Program 6–1 will plot the root-locus diagram for this system. The plot is shown in Figure 6–15.

MATLAB Program 6–1

```
% ---------- Root-locus plot ----------

num = [0   0   0   1   2   4];
den = [1   11.4   39   43.6   24   0];
rlocus(num,den)

Warning: Divide by zero
v = [-10   10   -10   10]; axis(v)
grid
title('Root-Locus Plot of G(s) = K(s^2 + 2s + 4)/[s(s + 4)(s + 6)(s^2 + 1.4s + 1)]')
```

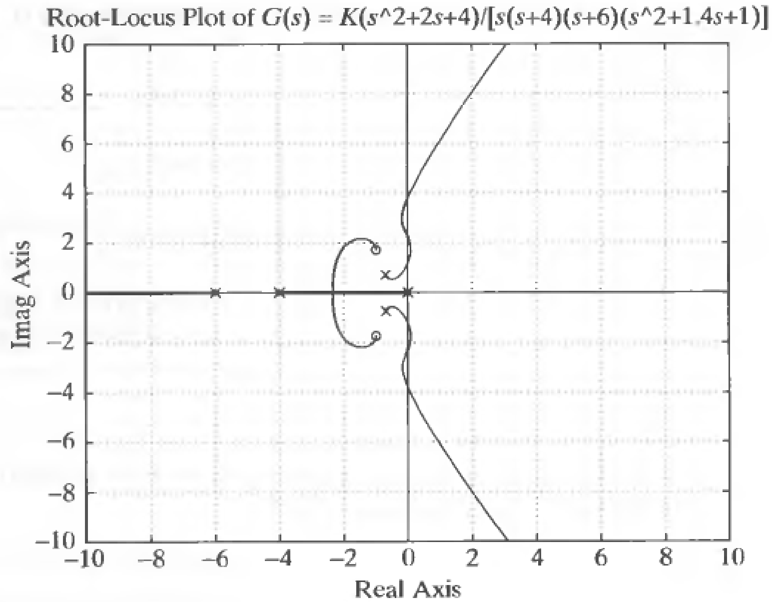


Figure 6–15
Root-locus plot.

EXAMPLE 6–4 Consider the system shown in Figure 6–16, where the open-loop transfer function $G(s)H(s)$ is

$$G(s)H(s) = \frac{K(s + 0.2)}{s^2(s + 3.6)}$$

The open-loop zero is at $s = -0.2$, and open-loop poles are at $s = 0, s = 0$, and $s = -3.6$.

MATLAB Program 6–2 generates a root-locus plot. The resulting root-locus plot is shown in Figure 6–17.

MATLAB Program 6–2

```
% ---------- Root-locus plot ----------

num = [0   0   1   0.2];
den = [1   3.6   0   0];
rlocus(num,den)
v = [-4   2   -4   4]; axis(v)
grid
title('Root-Locus Plot of G(s) = K(s + 0.2)/[s^2(s + 3.6)]')
```

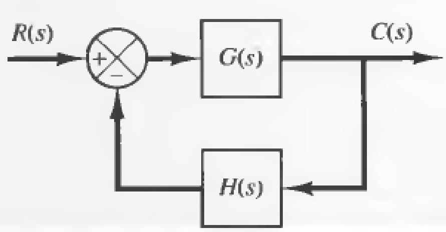


Figure 6–16
Control system.

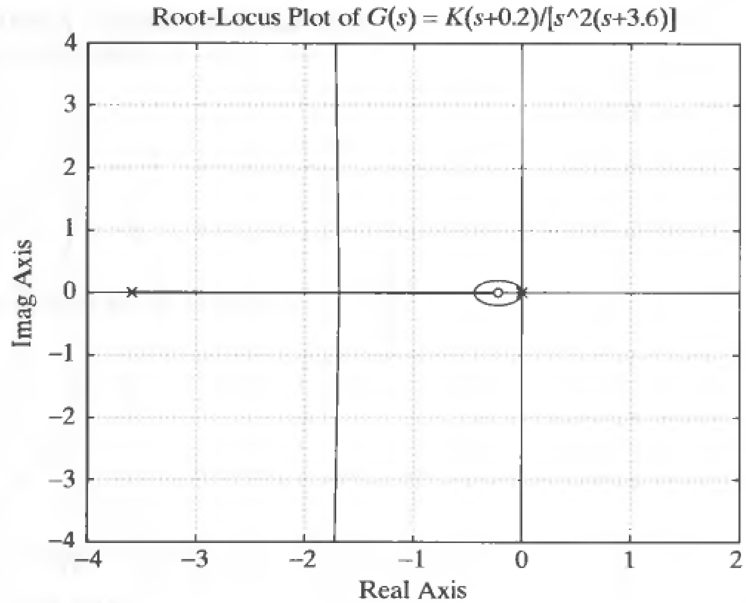


Figure 6–17
Root-locus plot.

EXAMPLE 6–5 Consider the system shown in Figure 6–18. Plot root loci with a square aspect ratio so that a line with slope 1 is a true 45° line.

To set the plot region on the screen to be square, enter the command axis('square'). With this command, a line with slope 1 is at a true 45°, not skewed by the irregular shape of the screen. (It is important to note that a hard-copy plot may or may not be of a square region depending on a printer.)

MATLAB Program 6–3 produces a root-locus plot in a square region. The resulting plot is shown in Figure 6–19.

MATLAB Program 6–3

```
% ---------- Root-locus plot ----------

num = [0   0   0   1   1];
den = [1   3   12   -16   0];
rlocus(num,den)
v = [-6   6   -6   6]; axis(v);axis('square')
grid
title('Root-Locus Plot of G(s) = K(s + 1)/[s(s - 1)(s^2 + 4s + 16)]')
```

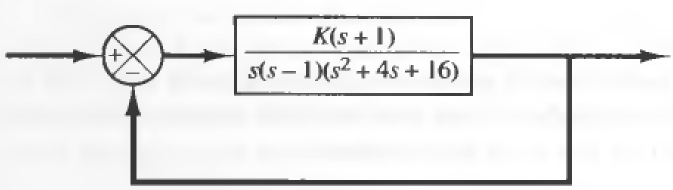


Figure 6–18
Control system.

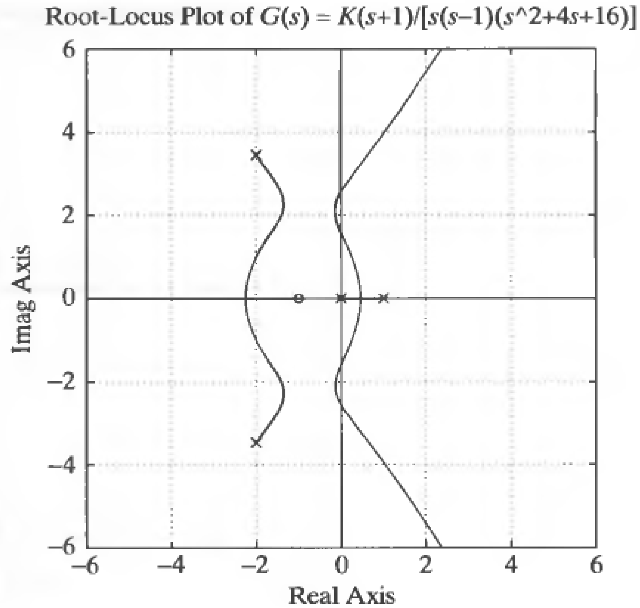


Figure 6–19
Root-locus plot.

EXAMPLE 6–6 Consider the system whose open-loop transfer function $G(s)H(s)$ is

$$G(s)H(s) = \frac{K}{s(s + 0.5)(s^2 + 0.6s + 10)}$$

$$= \frac{K}{s^4 + 1.1s^3 + 10.3s^2 + 5s}$$

There are no open-loop zeros. Open-loop poles are located at $s = -0.3 + j3.1480$, $s = -0.3 - j3.1480$, $s = -0.5$, and $s = 0$.

Entering MATLAB Program 6–4 into the computer, we obtain the root-locus plot shown in Figure 6–20.

MATLAB Program 6–4

```
% ---------- Root-locus plot ----------

num = [0   0   0   0   1];
den = [1   1.1   10.3   5   0];
rlocus(num,den)
grid
title('Root-Locus Plot of G(s) = K/[s(s + 0.5)(s^2 + 0.6s + 10)]')
```

Notice that in the regions near $x = -0.3$, $y = 2.3$ and $x = -0.3$, $y = -2.3$ two loci approach each other. We may wonder if these two branches should touch or not. To explore this situation, we may plot the root loci using the command

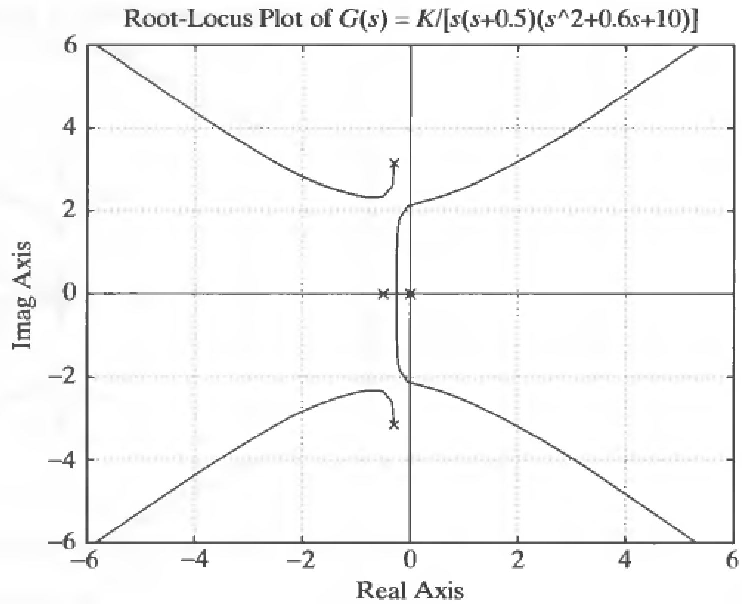


Figure 6–20
Root-locus plot.

r = rlocus(num,den)
plot(r,'o')

as shown in MATLAB Program 6–5. Figure 6–21 shows the resulting plot.

MATLAB Program 6–5

```
% ---------- Root-locus plot ----------

num = [0   0   0   0   1];
den = [1   1.1   10.3   5   0];
r = rlocus(num,den);
plot(r,'or')
v = [−6   6   −6   6]; axis(v)
grid
title('Root-Locus Plot of G(s) = K/[s(s + 0.5)(s^2 + 0.6s + 10)]')
xlabel('Real Axis')
ylabel('Imag Axis')

% ***** Note that the command 'plot(r,'or')' gives small circles
% in the screen plot in red color *****
```

Since there are no computed points near $(-0.3, 2.3)$ and $(-0.3, -2.3)$, it is necessary to adjust steps in gain K. By a trial and error approach, we find the particular region of interest to be $20 \le K \le 30$. By entering MATLAB Program 6–6, we obtain the root-locus plot shown in Figure 6–22. From this plot, it is clear that the two branches that approach in the upper half-plane (or in the lower half-plane) do not touch.